Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Tests

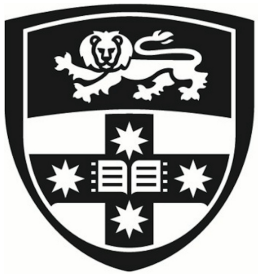
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Abstract

For two categorical variables, mutual information (MI) measures how much observing one reduces uncertainty about the other, on average. Independent samples from two populations can give different MI estimates even when the populations have equal MI. With small samples or rarely observed category combinations, the Wald test can reject equal MI too often or too rarely.

This thesis develops the Expanded Welch–Satterthwaite test (Expanded Welch) for comparing MI between two populations. It retains the estimated MI difference and standard error used by the Wald test but replaces the standard normal reference distribution with a Student t distribution. The standard error depends on variance estimates that change between samples. We derive a first-order approximation to how much these estimates vary and use it to estimate the degrees of freedom. The resulting Student t distribution gives the Expanded Welch test a larger rejection cutoff than that of the Wald test. Simulations under equal and unequal population MI assess approximation accuracy, false-positive rates, and power across sample sizes, numbers of categories, and category frequencies. An online-shopping example illustrates the tests on observed data.

In the simulations, the Expanded Welch test brings some excessive false-positive rates closer to the 5% target, but lowers rejection further where the Wald test already rejects too rarely. The larger cutoff can reduce power, the probability of detecting a real MI difference, although the power difference between methods becomes small with larger samples. The first-order approximation also becomes less accurate when the variables are nearly independent. In sparse settings, bias can reverse which population appears to have greater MI. The larger cutoff therefore improves some false-positive rates but cannot correct bias in the estimated MI difference or errors in its standard error.

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CHAPTER 1

Introduction

Mutual information (MI) measures the average reduction in uncertainty about one variable obtained by observing another (Bossomaier et al., 2016). Knowing one variable can provide partial information about the other, helping identify which outcomes are more likely without determining the outcome exactly. When the variables are independent, observing one provides no information about the other, and their MI is zero. Larger MI values indicate a greater average reduction in uncertainty.

The same pair of categorical variables X and Y can have different MI in different populations. This thesis develops and evaluates a statistical test for whether two populations have equal MI, using independently collected samples from each population.

Let P and Q be the joint probability distributions of X and Y in the two populations. The null and alternative hypotheses are

$$H_0 : I(P) = I(Q), \qquad H_1 : I(P) \neq I(Q). \tag{1.1}$$

In practice, MI is estimated from observed counts that vary from sample to sample, so two samples can produce different MI estimates even when the population MI values are equal. A statistical test must account for this sampling variation before an estimated difference can be interpreted as evidence of a population difference.

1.1 Why the comparison is difficult

The Wald test (Wald, 1943; van der Vaart, 1998) divides the estimated MI difference by its standard error and compares the resulting statistic with a standard normal distribution. At a 5% significance level, the intended false-positive rate is 5% when the population MI values are equal. The approximation using a

normal distribution can be inaccurate in small or sparse samples, causing the test to reject equal MI too often or too rarely.

Both the MI difference and its standard error are estimated from count tables, whose cells record how often each pair of categories occurs. MI depends on the cell probabilities and their row and column totals. When samples are small and rare combinations of categories occur only a few times or not at all, the resulting probability estimates can distort both the MI difference and its estimated uncertainty.

The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains the estimated MI difference and standard error used by the Wald test but uses a Student t distribution instead of the standard normal distribution. Its degrees of freedom depend on how the variance estimates underlying the standard error change between samples. This thesis derives a first-order approximation to that sampling variation, accounting for changes in cell probabilities and their row and column totals, and uses the result to obtain degrees of freedom for the Expanded Welch test.

The cutoff from the Student t distribution is larger than the Wald test's cutoff, so the Expanded Welch test's rejection rate cannot exceed that of the Wald test. A lower rejection rate can reduce false positives but can also reduce statistical power, the probability of rejecting equal MI when the population MI values genuinely differ. The evaluation compares false-positive rates across MI values shared by the two populations and power across MI differences. Both studies vary table sizes, category frequencies, and sample sizes to show where reduced false-positive rates accompany a loss of detection probability.

1.2 Research questions

The thesis addresses three research questions:

RQ1: How accurately do the derived approximations predict the mean and sampling variance of the estimated pointwise-MI variance, and what degrees of freedom do they imply near and away from independence between X and Y?

RQ2: How do the false-positive rates of the Expanded Welch test compare with those of the Wald test as shared population MI, sample size, table size, and category frequencies change?

RQ3: How does the power of the Expanded Welch test compare with that of the Wald test as the population MI difference, sample size, table size, and category frequencies change?

1.3 Contributions to the field

(1) **Mathematical derivation.** The thesis derives a first-order approximation to the sampling variability of the estimated pointwise-MI variance, accounting for changes in cell probabilities and their row and column totals.

(2) **Test construction.** The Expanded Welch test uses this approximation to estimate the degrees of freedom of a Student t reference distribution while retaining the estimated MI difference and standard error used by the Wald test.

(3) **Empirical evaluation.** Controlled simulations assess approximation accuracy and compare the tests' false-positive rates, power, and calculation success across sample sizes, table sizes, category frequencies, and MI values. The results identify where the larger cutoff brings false-positive rates closer to 5% or lowers rates already below 5%, and quantify the accompanying loss of power. The evaluation also shows how bias can reverse which population appears to have greater MI and how inaccurate standard errors can inflate or suppress rejection.

1.4 Scope and organisation

This thesis compares MI between two populations using categorical samples summarised in count tables with the same fixed row and column categories. The two samples are independent, and observations within each sample are also independent. For the main derivation, both populations are assumed to have positive MI and a positive probability in every cell. The case of zero MI, where X and Y are independent, is considered separately. The main simulations compare the Wald test and the Expanded Welch test using two-sided tests at the 5% significance level.

Chapter 2 reviews the background and literature. Chapters 3 and 4 define the statistic and derive the Expanded Welch test. Chapters 5 and 6 present the experiment, results, and online-shopping application. Chapter 7 brings together the findings, limitations, and future work. Appendix A gives supporting near-independence calculations, and Appendix B gives experimental construction, calculation requirements, and simulation uncertainty details.

CHAPTER 2

Background and Literature Review

Testing for equal MI in two populations requires an estimate from each sample and its sampling variance. This chapter develops the MI estimator, derives its variance, and reviews tests for equal population MI.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

Entropy measures uncertainty about a variable's value, and MI measures how much that uncertainty is reduced, on average, by observing another variable (Shannon, 1948).

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. The Shannon entropy is the expected information content, measured here in nats using natural logarithms:

$$H(X) = -\sum_{x} p_X(x) \log p_X(x). \tag{2.1}$$

For a pair (X, Y), the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X \mid Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y.

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \ldots, r\}$ and $Y \in \{1, \ldots, c\}$. The numbers of categories, r and c, are their alphabet sizes. Write

$$p_{i+} = \sum_{j=1}^{c} p_{ij}, \qquad p_{+j} = \sum_{i=1}^{r} p_{ij} \tag{2.2}$$

for the marginal probabilities, or row and column margins. The pointwise mutual information (also called local MI) associated with cell (i, j) is

$$\ell_P(i, j) = \log \left(\frac{p_{ij}}{p_{i+} p_{+j}} \right). \tag{2.3}$$

The denominator is the cell probability under independence with the same margins. Positive pointwise MI means that the cell occurs more often than independence predicts, whereas negative pointwise MI means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^{r} \sum_{j=1}^{c} p_{ij} \ell_P(i, j). \tag{2.4}$$

Here the summations are nested: for each row i, the inner sum runs over all columns j. Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$.

Because P is the joint distribution of (X, Y), we also write $I(P)$ as $I(X; Y)$. Its entropy form is

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \tag{2.5}$$

MI is therefore the average reduction in uncertainty about X obtained by observing Y, or equivalently about Y obtained by observing X. It is nonnegative and equals zero exactly when $p_{ij} = p_{i+} p_{+j}$ for every cell, including zero-probability cells (Bossomaier et al., 2016).

Although MI measures the overall departure from independence, it does not specify the shape of association. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions, so the null hypothesis requires equal MI without requiring identical cell probabilities in the two population tables.

The logarithm base determines the reporting unit, and this thesis uses natural logarithms to express MI in nats. Dividing by $\log 2$ converts MI to bits by rescaling MI and its standard error together, while leaving the degrees of freedom unchanged. The test result is therefore identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij}. The empirical cell probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}}{n}. \tag{2.6}$$

The plug-in estimator inserts these observed relative frequencies into Equation (2.4):

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij} \log \left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}} \right). \tag{2.7}$$

The zero-cell convention follows from $\lim_{x\downarrow 0} x \log x = 0$, so empty cells contribute no term to the sum. When a positive-probability cell is unobserved, the empirical table assigns it zero probability, contributing to sparse-sample estimation error.

When the counts follow a multinomial distribution, they vary between samples but must sum to n, so an extra observation in one cell leaves fewer observations for the others. For a fixed set of categories, the plug-in estimate approaches population MI as the sample grows, but its finite-sample bias and sampling variance depend on the complete probability table. Changing the probabilities of rare cells can change both the bias and the sampling variance even when the categories and sample size are unchanged (Paninski, 2003).

Bias is the difference between the average estimate across repeated samples and the population value: $\mathrm{Bias}\{\widehat{I}(P)\} = \mathrm{E}\{\widehat{I}(P)\} - I(P)$. Even when the population variables are independent, sampling fluctuations can produce a table with apparent association and positive plug-in MI. These estimates cannot be offset by negative plug-in MI values, so their average can exceed the true value of zero. For a fixed $r \times c$ table with positive population cell probabilities, the large-sample expansion has a positive leading term:

$$\mathrm{Bias}\{\widehat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \tag{2.8}$$

The remainder $O(n^{-2})$ collects terms bounded in magnitude by a constant times n^{-2} as n grows for this fixed population table. The leading term grows with table size and shrinks as the sample grows, but small or sparse samples can have further error beyond this approximation (Paninski, 2003). The result is

closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989).

When $I(P) = I(Q)$ but $n_P \neq n_Q$, unequal sample sizes give the uncorrected MI difference a leading bias of

$$\frac{(r-1)(c-1)}{2}\left(\frac{1}{n_P} - \frac{1}{n_Q}\right). \tag{2.9}$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MI values. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

Estimation beyond the leading bias correction.

Data are undersampled when too few observations are available to estimate the joint probabilities reliably, which can leave rare category combinations poorly represented or absent from the sample. Panzeri and Treves (1996) derive analytical bias corrections that account for the categories with nonzero population probability, but these remain large-sample approximations. Paninski (2003) shows that substantial error can remain after standard corrections, including the jackknife, when many categories are poorly sampled. To account for categories missing from the sample, Chao and Shen (2003) adjust entropy using sample coverage, the total population probability of the categories that have been observed. Their adjustment also accounts for the different probabilities of discovering common and rare categories.

Bayesian estimators model the unknown probabilities rather than treating observed proportions as the complete population. The Nemenman–Shafee–Bialek (NSB) estimator combines priors, or probability weights on possible populations, to avoid favouring a narrow range of entropy values before observing the data (Nemenman et al., 2002). More recently, Piga et al. (2024) use the observed counts to estimate the strength of a Dirichlet prior distribution, which controls how much the probability estimates are drawn towards a uniform distribution, including for unobserved categories.

Entropy estimators can also be combined to estimate MI through $I(X;Y) = H(X) + H(Y) - H(X,Y)$, as in Equation (2.5). Archer et al. (2013) study this route alongside estimators that model the joint probabilities directly. Combining three separately constructed Bayesian entropy estimates need not correspond to one joint probability model and can produce negative MI estimates. Nevertheless, these combined estimators generally outperform the authors' fully Bayesian MI estimator in their tested examples.

Hernández and Samengo (2019) tailor MI estimation to a well-sampled, low-entropy variable paired with a variable whose many categories are poorly sampled. Their estimator uses the well-estimated marginal and models the conditional probabilities, rather than estimating three entropies separately. Their results also show that combining Bayesian entropy estimates can retain bias under severe undersampling. When MI estimates from two distinct populations are subtracted, their estimation errors need not cancel, so the two-sample problem requires both a small bias in the estimated difference and a standard error that describes its sampling variation.

2.3 Sampling variance of plug-in MI

Two MI estimates can differ even when their population values are equal, because each estimate varies between samples. Testing their difference therefore requires a standard error that accounts for this sampling variation. We first derive the established large-sample approximation for the variance of one plug-in MI estimate:

$$\mathrm{Var}\{\widehat{I}(P)\} \approx \frac{V(P)}{n}. \tag{2.10}$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004). Section 2.4 then explains how this result supports a comparison between two populations.

The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_{i} p_{i+} \log p_{i+} - \sum_{j} p_{+j} \log p_{+j}. \tag{2.11}$$

Each sum contains the same function, $f(u) = u \log u$. Expanding this function lets us translate probability errors into changes in MI. The derivative needed for its first-order Taylor expansion follows directly from

the product rule:

$$f'(u) = \frac{d}{du}\{u \log u\} = \log u + u\frac{1}{u} = \log u + 1. \tag{2.12}$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \tag{2.13}$$

where a is the value around which the function is expanded and $x - a$ is the distance from that value. Here the new value is the estimated probability $x = \widehat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \widehat{u} - u$, which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\widehat{u} \log \widehat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\widehat{u} - u)}_{\text{first-order change}}. \tag{2.14}$$

Subtracting the population value from both sides makes the change explicit:

$$\widehat{u} \log \widehat{u} - u \log u \approx (\log u + 1)(\widehat{u} - u). \tag{2.15}$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \widehat{I}(P) - I(P) \approx & \sum_{i,j} (\log p_{ij} + 1)(\widehat{p}_{ij} - p_{ij}) \\ & - \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) \\ & - \sum_{j} (\log p_{+j} + 1)(\widehat{p}_{+j} - p_{+j}). \end{aligned} \tag{2.16}$$

The margin errors are sums of the corresponding cell errors:

$$\widehat{p}_{i+} - p_{i+} = \sum_{j} (\widehat{p}_{ij} - p_{ij}), \qquad \widehat{p}_{+j} - p_{+j} = \sum_{i} (\widehat{p}_{ij} - p_{ij}). \tag{2.17}$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) &= \sum_{i} \left[(\log p_{i+} + 1) \left\{ \sum_{j} (\widehat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.18}$$

The inner sum over j is the probability error for a fixed row i. It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way.

After this rearrangement, the same cell error $\widehat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term, and its column-margin term. Its combined coefficient is

$$\begin{aligned}
&(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\
&\quad = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right) - 1 \\
&\quad = \ell_P(i,j) - 1. \qquad (2.19)
\end{aligned}$$

Therefore,

$$\widehat{I}(P) - I(P) \approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}). \qquad (2.20)$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j} (\widehat{p}_{ij} - p_{ij}) = \sum_{i,j} \widehat{p}_{ij} - \sum_{i,j} p_{ij} = 1 - 1 = 0. \qquad (2.21)$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned}
\widehat{I}(P) - I(P) &\approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}) \\
&= \sum_{i,j} \ell_P(i,j)(\widehat{p}_{ij} - p_{ij}) - \sum_{i,j} (\widehat{p}_{ij} - p_{ij}) \\
&= \sum_{i,j} \ell_P(i,j)(\widehat{p}_{ij} - p_{ij}). \qquad (2.22)
\end{aligned}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a. An empirical cell probability is the proportion of observations in that cell, so

$$\widehat{p}_{ij} - p_{ij} = \frac{1}{n} \sum_{a=1}^{n} \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}. \qquad (2.23)$$

Substituting this expression into Equation (2.22) first gives

$$\widehat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i,j) \left[\frac{1}{n} \sum_{a=1}^{n} \{ \mathbf{1}(Z_a = (i,j)) - p_{ij} \} \right]. \tag{2.24}$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned} \widehat{I}(P) - I(P) &\approx \frac{1}{n} \sum_{a=1}^{n} \left[\sum_{i,j} \ell_P(i,j) \mathbf{1}(Z_a = (i,j)) - \sum_{i,j} p_{ij} \ell_P(i,j) \right] \\ &= \frac{1}{n} \sum_{a=1}^{n} \{ \ell_P(Z_a) - I(P) \}. \end{aligned} \tag{2.25}$$

The first sum inside the brackets selects the single cell containing Z_a, giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$.

To first order, Equation (2.25) expresses the MI error as an average of n observation-level contributions. The contribution of an observation in cell (i,j) is simply that cell's pointwise MI minus the population average, $\ell_P(i,j) - I(P)$. Its mean is zero because

$$\begin{aligned} \mathrm{E}_P\{\ell_P(X,Y) - I(P)\} &= \sum_{i,j} p_{ij} \{ \ell_P(i,j) - I(P) \} \\ &= \underbrace{\sum_{i,j} p_{ij} \ell_P(i,j)}_{I(P)} - I(P) \underbrace{\sum_{i,j} p_{ij}}_{1} = 0. \end{aligned} \tag{2.26}$$

The first-order term in the MI error therefore has mean zero. The higher-order terms omitted by this approximation generate the leading bias described in Section 2.2. The centred function $\ell_P(i,j) - I(P)$ is called the influence function of MI. Equation (2.25) uses it to approximate the sampling error (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y) - I(P)\} = \sum_{i,j} p_{ij} \{ \ell_P(i,j) - I(P) \}^2. \tag{2.27}$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between the contributions of any two distinct observations is therefore zero. Hence

$$\begin{aligned}\mathrm{Var}\{\widehat{I}(P)\} &\approx \mathrm{Var}\left\{\frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}\mathrm{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \tag{2.28}$$

Averaging n independent contributions therefore divides their variance by n. For dependent observations, covariance terms must also be included in Equation (2.28), as studied by Moddemeijer (1999). The present experiment uses independent observations.

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\widehat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0,1). \tag{2.29}$$

This gives a large-sample approximation using a standard normal distribution for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i,j) = \log 1 = 0$ in every cell. Hence $I(P) = 0, V(P) = 0$, and the first-order average in Equation (2.25) vanishes. The plug-in MI estimate can still be positive when population MI is zero. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limiting distribution (Brillinger, 2004; Marinescu and Balcau, 2025). Testing independence therefore uses a chi-squared distribution rather than an approximation using a normal distribution for comparing positive MI values.

2.4 Approaches to testing MI differences

For two independent samples, the variance contributions described in the preceding section add to approximate the sampling variance of the MI difference. Estimating this sum and taking its square root gives the standard error, which estimates the difference's standard deviation across repeated samples.

This section reviews the Wald test, the Welch–Satterthwaite principle, and resampling alternatives. The hypothesis is that the populations have the same MI, $I(P) = I(Q)$.

The Wald test.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with a standard normal distribution (Wald, 1943; van der Vaart, 1998). For MI estimated from independent samples from P and Q, the first-order variance contributions add:

$$\operatorname{Var}\{\widehat{I}(P) - \widehat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \tag{2.30}$$

Each term approximates the sampling variance of one MI estimate, with independence of the two samples making their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\widehat{I}(P) - \widehat{I}(Q)}{\sqrt{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q}} \xrightarrow{d} N(0,1) \qquad \text{under } I(P) = I(Q). \tag{2.31}$$

Here $\widehat{V}(P)$ and $\widehat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. The approximation using a standard normal distribution assumes positive population cell probabilities, positive pointwise-MI variances, and increasing sample sizes. With smaller samples, a normal distribution may approximate the MI difference poorly, and the estimated standard error can vary between samples. The Wald test still uses the same standard normal distribution cutoff, regardless of how much the estimated standard error varies. Chapter 3 applies the leading bias correction to the numerator and specifies the full Wald test.

The Welch–Satterthwaite principle.

This thesis adapts the Welch–Satterthwaite principle from comparing population means to comparing population MI values. Welch's test allows the population variances to differ and compares the standardised difference between two sample means with a Student t distribution with ν degrees of freedom (Welch, 1947). Here $\nu > 0$ controls the distribution's tail weight; an effective value need not be an integer. For two independent samples with sample means $\bar{X}_P, \bar{X}_Q$ and sample variances s_P^2, s_Q^2, the statistic for testing equal population means and the effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \tag{2.32}$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\dfrac{(s_P^2/n_P)^2}{n_P - 1} + \dfrac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \tag{2.33}$$

The denominator contains one estimated variance contribution from each sample, with component degrees of freedom $n_P - 1$ and $n_Q - 1$ under sampling from populations with normal distributions. If s_P^2/n_P dominates, the formula makes the combined ν close to $n_P - 1$; when both contributions matter, both samples' degrees of freedom affect ν. Satterthwaite obtains this form by matching the mean and variance of their sum to those of a variable with a scaled chi-squared distribution (Satterthwaite, 1946).

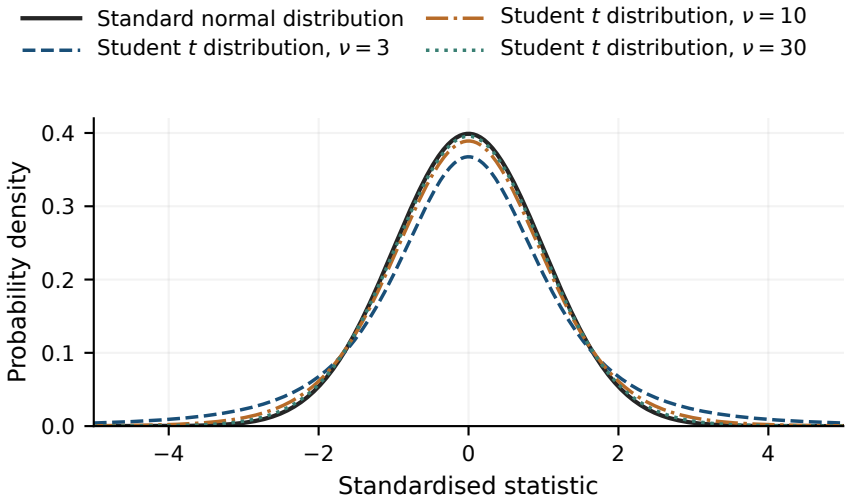


FIGURE 2.1: Standard normal distribution and Student t distributions with 3, 10, and 30 degrees of freedom.

Figure 2.1 shows that the Student t distribution has heavier tails than the standard normal distribution, especially when ν is small, so a larger absolute statistic is needed to reject the null at the same significance level. As ν increases, the Student t distribution approaches the standard normal distribution, and their rejection cutoffs become closer.

For MI, the estimated variance contributions are $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. The ordinary $n_P - 1$ and $n_Q - 1$ values follow from the scaled chi-squared distribution of a sample variance when each population has a normal distribution; they do not automatically apply to an estimated pointwise-MI variance. Chapter 4 therefore derives the component degrees of freedom from the sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$, using the same moment-matching principle.

The estimated MI sampling variance $\widehat{V}(P)/n_P$ describes uncertainty in $\widehat{I}(P)$, whereas $\mathrm{Var}\{\widehat{V}(P)\}$ measures how much the variance estimate itself changes between samples. Even variance estimators with the same mean can have different sampling variances, so Satterthwaite's calculation matches both moments. Greater sampling variance relative to the squared mean gives fewer degrees of freedom and a larger rejection cutoff (Satterthwaite, 1946). A sample-size assignment alone does not adapt to differences in this relative variability between probability tables.

The same principle has been used in regression to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001; Imbens and Kolesár, 2016). Pan and Wall (2002) use Student t distributions and F distributions with approximate degrees of freedom that account for the sampling variability of a regression variance estimate to reduce excessive Wald rejection. Bell and McCaffrey (2002) examine both bias reduction in the variance estimator and Satterthwaite degrees of freedom, finding that each contributes to improved inference. Their experiments also show that using Satterthwaite degrees of freedom with a positively biased variance estimator can produce very conservative tests. The distinction matters for MI: a larger cutoff can compensate for some excessive rejection, but it does not change the estimated variance or remove bias in the estimated MI difference.

Resampling alternatives.

A bootstrap approximates sampling variation by repeatedly drawing samples with replacement from the observed data (Efron, 1979). For this comparison, samples would be drawn separately from the two observed tables, using their original sample sizes, and the MI difference recalculated for each pair. Resampling from separate empirical probability tables allows their cell probabilities to differ and provides an estimated sampling distribution of the MI difference. The bootstrap variation around the observed estimate could then be used to construct an uncertainty interval for the population difference and assess whether that interval includes zero.

Because an ordinary empirical bootstrap assigns an unobserved cell zero probability, it cannot sample that cell and can therefore miss error arising from unobserved cells. Paninski (2003) gives a broader warning: confidence intervals based on resampling can be too narrow when information estimates remain biased.

A simple permutation test combines the two samples and shuffles their group labels, giving an exact test when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only $I(P) = I(Q)$, allowing the tables to differ while their MI values are equal. Equal MI alone therefore does

not guarantee that this label-permutation test has the stated false-positive rate (Chung and Romano, 2013). Although specialised permutation methods allow distinct populations in some parameter comparisons (Janssen, 1997; Chung and Romano, 2013), their validity for equal MI would need to be established.

2.5 Previous work and this thesis's contribution

Existing uncertainty calculations describe how much an MI estimate varies between samples. Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. The first-order results of Moddemeijer (1989) and Brillinger (2004) give the MI sampling variance as approximately $V(P)/n_P$, as developed in Section 2.3.

In a segregation application, Mora and Ruiz-Castillo (2009) derive an approximation using a normal distribution for comparing positive population MI between populations and time periods, providing a direct precedent for the Wald test used as the baseline. Their later peer-reviewed account places the MI index within the segregation literature (Mora and Ruiz-Castillo, 2011).

Hutcheson's test provides a related precedent for comparing estimated information quantities using a Student t distribution as an approximate reference. It compares two estimates of Shannon diversity, or entropy, using a Welch-type statistic with effective degrees of freedom

$$\nu_H = \frac{(s^2_{H,P} + s^2_{H,Q})^2}{(s^2_{H,P})^2/n_P + (s^2_{H,Q})^2/n_Q} \tag{2.34}$$

(Hutcheson, 1970). Here each of $s^2_{H,P}$ and $s^2_{H,Q}$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$. The formula assigns the two variance components degrees of freedom n_P and n_Q, rather than the $n_P - 1$ and $n_Q - 1$ values used in the ordinary Welch test.

For MI, the row, column, and joint entropies are calculated from the same observations, so their estimation errors are connected through the cell counts contributing to all three. The MI sampling-variance formulas in Section 2.3 account for this dependence when estimating uncertainty in MI.

In a Bayesian approach to uncertainty about population MI, Hutter (2002) and Hutter and Zaffalon (2005) model the unknown cell probabilities and calculate the variance and shape of the resulting MI distribution conditional on the observed table. Our repeated-sampling comparison instead asks how often the Wald test and the Expanded Welch test reject equal MI when new samples are drawn from fixed populations P and Q.

MI has also been used to test whether complete population distributions are equal. Guha and Chothia (2014) compare two continuous distributions using MI between pooled observations and their sample labels, while Drake and Guha (2014) develop an MI-based k-sample test for discrete data. This thesis compares the MI of two variables within each population, testing $I(P) = I(Q)$.

The additional calculation needed by the Expanded Welch test concerns how much the estimated pointwise-MI variance changes between samples. The moment-matching rule in Section 2.4 requires both the sampling mean and variance of this estimate to obtain component degrees of freedom.

This thesis derives a first-order approximation to that sampling variance by tracking how changes in observed counts affect the estimated pointwise-MI variance. A change in a cell count alters both the estimated probability weights and the pointwise-MI values calculated from the joint and marginal probabilities. Following the moment-matching approach used in regression by Pan and Wall (2002), the Expanded Welch test uses this sampling-variance approximation to obtain component degrees of freedom. The component degrees of freedom are then combined using the Welch–Satterthwaite rule.

The Wald test and the Expanded Welch test use the same estimated MI difference and standard error, so their comparison isolates the effect of changing the reference distribution. The estimation problems identified by Paninski (2003) remain relevant: residual bias can shift the estimated difference. For a fixed estimated difference, an underestimated standard error can make the absolute test statistic too large, while an overestimated standard error can make it too small. The evaluation therefore checks the variance-moment approximations separately from false-positive rates and power.

CHAPTER 3

Overview of the Two-Sample MI Test and Wald Baseline

This chapter defines the estimated MI difference, standard error, and test statistic shared by the Wald test and the Expanded Welch test, then calculates the Wald p-value using the standard normal distribution.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \qquad N^Q = (N_{ij}^Q) \tag{3.1}$$

be independent $r \times c$ count tables with totals n_P and n_Q. The rows and columns represent the same categorical variables X and Y in both populations, with the same category definitions. The empirical probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \qquad \widehat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \tag{3.2}$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \tag{3.3}$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \tag{3.4}$$

The large-sample argument uses the following assumptions.

ASSUMPTION 1 (Sampling). *Observations are independent and identically distributed within each population, and the two population samples are independent.*

ASSUMPTION 2 (Fixed categories shared by both tables). *The row and column categories are fixed in advance and aligned between the two tables.*

ASSUMPTION 3 (Positive cell probabilities). *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare, and we retain such samples.*

ASSUMPTION 4 (Positive MI). *Both populations have positive MI:* $I(P) > 0$ *and* $I(Q) > 0$*. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive pointwise-MI variances* $V(P) > 0$ *and* $V(Q) > 0$*. The positive-MI assumption excludes exact independence, where the first-order approximation breaks down.*

Appendix A.3 explains this equivalence and why positive cell probabilities are required.

All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats.

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\widehat{\ell}_P(i,j) = \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right) \qquad \text{when } \widehat{p}_{ij} > 0. \tag{3.5}$$

The plug-in MI estimate is

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\widehat{\ell}_P(i,j). \tag{3.6}$$

The same calculation gives $\widehat{I}(Q)$.

For a fixed set of row and column categories, subtracting the leading MI bias in Equation (2.8) (Miller, 1955; Moddemeijer, 1989) from each plug-in estimate gives the bias-corrected MI difference:

$$\widehat{\Delta}_{\mathrm{BC}} = \left\{\widehat{I}(P) - \frac{(r-1)(c-1)}{2n_P}\right\} - \left\{\widehat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q}\right\}. \tag{3.7}$$

The two terms in braces are the bias-corrected MI estimates for the separate populations. Their corrections use the declared row and column categories because every population cell has positive probability under the model, even when some cells are unobserved in a sample. We retain negative corrected estimates when calculating the difference. When $n_P = n_Q$, the two corrections cancel, so $\widehat{\Delta}_{\mathrm{BC}}$ is exactly the raw plug-in MI difference. With unequal sample sizes, the difference between the two corrections removes the leading unequal-sample bias described in Section 2.2.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\widehat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates the pointwise-MI variance $V(P)$ by

$$\widehat{V}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2. \tag{3.8}$$

Define $\widehat{V}(Q)$ analogously. These estimates describe variation in pointwise MI across observation outcomes, and dividing each by its sample size estimates the sampling variance of the corresponding MI estimate.

Independence of the two samples makes the covariance between their MI estimates zero, so we estimate the sampling variance of their difference by adding the two estimated variance contributions:

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}. \tag{3.9}$$

Taking the positive square root gives $\widehat{\mathrm{SE}}$, which estimates the standard deviation of $\widehat{\Delta}_{\mathrm{BC}}$ across repeated samples. Table 3.1 distinguishes the variation of pointwise MI across observation outcomes from the sampling variation of the estimates calculated from whole tables.

TABLE 3.1: Variance quantities in the two-sample MI test.

Quantity	**Meaning**
$V(P)$	Population variance of pointwise MI; $V(P)/n_P$ approximates $\mathrm{Var}\{\widehat{I}(P)\}$.
$\widehat{V}(P)$	Estimate of $V(P)$ from one sampled count table.
$\widehat{V}(P)/n_P$	Estimated sampling variance of $\widehat{I}(P)$; one contribution to $\widehat{\mathrm{SE}}^2$.
$\mathrm{Var}\{\widehat{V}(P)\}$	Sampling variance of $\widehat{V}(P)$ across repeated samples; used to choose the component degrees of freedom.
$\widehat{\mathrm{SE}}^2$	Estimate of $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, calculated as $\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q$.
$\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$	Sampling variance of $\widehat{\Delta}_{\mathrm{BC}}$; the quantity that $\widehat{\mathrm{SE}}^2$ estimates.

The shared standardised statistic is

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}. \tag{3.10}$$

The sign of T indicates whether P or Q has the larger corrected MI estimate, while its magnitude measures the difference relative to its estimated standard error. Both tests use the same effect estimate and plug-in standard error in Equations (3.7)–(3.10), but compare T with different reference distributions centred at zero under H_0.

3.3 Wald test and motivation for the Expanded Welch test

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0,1) \qquad \text{under } H_0. \tag{3.11}$$

The two-sided Wald p-value is

$$p_{\mathrm{Wald}} = 2\{1 - \Phi(|T|)\}, \tag{3.12}$$

where Φ is the standard normal distribution function. Under this reference distribution for equal MI, the p-value is the probability of a statistic at least as far from zero as the observed T. The two-sided calculation doubles the upper-tail probability $1 - \Phi(|T|)$ because symmetry gives the same probability in the lower tail. At the nominal 5% level, the test rejects when $p_{\mathrm{Wald}} \leq 0.05$.

Why use a Student t distribution?

The large-sample limit above relies on $\widehat{\mathrm{SE}}$ estimating the sampling standard deviation of $\widehat{\Delta}_{\mathrm{BC}}$ with decreasing relative error as sample sizes increase. With smaller samples, variation in $\widehat{\mathrm{SE}}$ can affect the distribution of the ratio T. For a fixed estimated MI difference, a smaller $\widehat{\mathrm{SE}}$ gives a larger $|T|$ and can move it beyond the rejection cutoff. The Wald test uses the same cutoff from the standard normal distribution regardless of how much $\widehat{V}(P)$ and $\widehat{V}(Q)$ vary between samples.

The Expanded Welch test uses a Student t distribution as an approximate reference for T. Its heavier tails give a larger rejection cutoff than the cutoff from the standard normal distribution at the same significance

level, with fewer degrees of freedom giving a larger adjustment. Under the Welch–Satterthwaite moment-matching rule, an estimated variance contribution with greater sampling variance relative to its squared sampling mean receives fewer component degrees of freedom (Satterthwaite, 1946).

We therefore need the average value of each estimated pointwise-MI variance and how much that estimate varies across samples. For P, to first order,

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P). \tag{3.13}$$

For the remaining input, Chapter 4 derives a first-order approximation to $\mathrm{Var}\{\widehat{V}(P)\}$ by tracking how the estimated pointwise-MI variance changes when one observation changes the joint table.

CHAPTER 4

Detailed Derivation of the Expanded Welch–Satterthwaite Test for MI

The Expanded Welch test keeps the estimated MI difference, standard error, and test statistic from Chapter 3, but changes the distribution used to judge that statistic from the standard normal distribution to a Student t distribution. To choose its degrees of freedom, this chapter calculates how much the estimated pointwise-MI variance changes between samples. The name Expanded Welch refers to this expansion of the pointwise-MI variance as the table probabilities change.

The derivation first expresses the degrees of freedom in terms of the sampling mean and variance of the estimated variance contributions. To approximate those moments, we calculate how one observation changes the variance estimate, express the sampling error as an average of these contributions, and take the variance of that average.

4.1 Choosing degrees of freedom for the Expanded Welch test

To connect the estimated standard error to the reference distribution, divide both the estimated MI difference and its standard error by the population sampling standard deviation of $\widehat{\Delta}_{\mathrm{BC}}$. This rewrites the shared statistic without changing its value:

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}} / \sqrt{\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})}}{\sqrt{\widehat{\mathrm{SE}}^2 / \mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})}}.$$

The squared denominator is the estimated sampling variance divided by the population sampling variance. This ratio is the variance term that the Student t construction below models.

The exact construction uses a numerator with a standard normal distribution and an independent variance term. If Z has a standard normal distribution and U, independently of Z, has a chi-squared distribution with ν degrees of freedom, the scaled ratio follows a Student t distribution:

$$\frac{Z}{\sqrt{U/\nu}} \sim t_\nu. \tag{4.1}$$

Under equal MI, $\widehat{\Delta}_{\mathrm{BC}}/\sqrt{\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})}$ approaches a standard normal distribution, as described in Chapter 3, so it plays the role of Z in the approximation. The normalised squared standard error $\widehat{\mathrm{SE}}^2/\,\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$ plays the role of U/ν, which has a scaled chi-squared distribution. For MI, we model the estimated variance contributions using scaled chi-squared distributions with matching sampling means and variances. Because moment matching does not establish independence between the MI numerator and the variance estimates, we use the Student t distribution as an approximate reference.

Moment matching for one pointwise-MI variance estimate.

The squared standard error adds $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. We first match moments for $\widehat{V}(P)$, then show why dividing by n_P leaves its component degrees of freedom unchanged. To obtain the component target $\nu_{P,\mathrm{mom}}$, Satterthwaite matches the mean and variance of the nonnegative estimate $\widehat{V}(P)$ with those of a variable following a scaled chi-squared distribution:

$$\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\mathrm{mom}}}}{\nu_{P,\mathrm{mom}}}. \tag{4.2}$$

A chi-squared distribution with $\nu_{P,\mathrm{mom}}$ degrees of freedom has moments:

$$\mathrm{E}(\chi^2_{\nu_{P,\mathrm{mom}}}) = \nu_{P,\mathrm{mom}}, \qquad \mathrm{Var}(\chi^2_{\nu_{P,\mathrm{mom}}}) = 2\nu_{P,\mathrm{mom}}. \tag{4.3}$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$\mathrm{E}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\mathrm{mom}}}}{\nu_{P,\mathrm{mom}}}\right] = \mathrm{E}\{\widehat{V}(P)\}. \tag{4.4}$$

The variance of the approximating variable is

$$\mathrm{Var}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\mathrm{mom}}}}{\nu_{P,\mathrm{mom}}}\right] = \frac{[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu^2_{P,\mathrm{mom}}}\,\mathrm{Var}(\chi^2_{\nu_{P,\mathrm{mom}}}) \tag{4.5}$$

$$= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_{P,\mathrm{mom}}}. \tag{4.6}$$

We now choose $\nu_{P,\mathrm{mom}}$ so that the approximating variable also has the same variance as $\widehat{V}(P)$. Equating its variance, calculated above, to the sampling variance of $\widehat{V}(P)$ requires

$$\frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_{P,\mathrm{mom}}} = \mathrm{Var}\{\widehat{V}(P)\}.$$

This condition determines the degrees of freedom for the approximating distribution. Solving for $\nu_{P,\mathrm{mom}}$ gives

$$\boxed{\nu_{P,\mathrm{mom}} = \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}}. \tag{4.7}$$

The squared standard error contains $\widehat{V}(P)/n_P$, so we apply moment matching to this contribution as well as to $\widehat{V}(P)$. Because the sample size n_P is fixed, its reciprocal can be taken outside the expectation and its square outside the variance. Applying the same moment-matching calculation to this variance contribution therefore gives

$$\begin{aligned}\frac{2[\mathrm{E}\{\widehat{V}(P)/n_P\}]^2}{\mathrm{Var}\{\widehat{V}(P)/n_P\}} &= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2/n_P^2}{\mathrm{Var}\{\widehat{V}(P)\}/n_P^2} \\ &= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}} = \nu_{P,\mathrm{mom}}.\end{aligned}$$

The factors $1/n_P^2$ cancel, so dividing the estimate by n_P leaves its moment-matched degrees of freedom unchanged.

The same calculation applies to the contribution from Q. Once both component inputs are available, they must be combined to choose one reference distribution for T. For component inputs ν_P and ν_Q, the Welch–Satterthwaite equation is (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\nu_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\nu_Q}. \tag{4.8}$$

Section 4.3 derives this rule by matching the mean and variance of the sum of the two independent variance contributions.

To evaluate Equation (4.7), we need the mean and variance of $\widehat{V}(P)$ across repeated samples from P. The first-order expansion derived in Section 4.3 gives

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P).$$

At this order, the change in $\widehat{V}(P)$ is linear in the empirical cell-probability errors. Each empirical cell proportion has expectation equal to its population probability, so these errors and their linear contribution have mean zero. Any bias therefore arises from terms beyond this linear approximation.

With the mean approximated by $V(P)$, the remaining task is to calculate $\mathrm{Var}\{\widehat{V}(P)\}$. Section 2.3 obtained the sampling variance of MI by writing its estimation error as an average of observation-level contributions. We use the same strategy for the pointwise-MI variance estimate: calculate the contribution of one observation to its error, then find the variance of the average. The next section derives that contribution, and Section 4.3 shows how averaging it approximates the error in $\widehat{V}(P)$ across whole sampled tables.

4.2 How pointwise-MI variance changes with the table

To find the contribution of one observation, we calculate how a small change in the probability table affects

$$V(P) = \mathrm{Var}_P\{\ell_P(X, Y)\}. \tag{4.9}$$

Writing this variance in second-moment form identifies the quantities whose changes we must calculate:

$$V(P) = \mathrm{E}_P\{\ell_P(X, Y)^2\} - I(P)^2. \tag{4.10}$$

The first term changes when either the probability weights or the pointwise-MI values change, and the second changes with MI. We will differentiate these terms along a path representing one observation.

Fix a cell (x, y) and define a distribution that moves a small amount of probability toward this cell (Hampel, 1974; van der Vaart, 1998):

$$P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_{(x,y)}, \tag{4.11}$$

where $\delta_{(x,y)}$ places probability one on cell (x, y). At $\varepsilon = 0$, the distribution is P. Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one. The direction $\delta_{(x,y)} - P$ is the difference between the table representing one observation and the population table, so it describes that observation's contribution to the empirical-table error.

The influence function of $V(P)$ measures its observation-level sensitivity and is defined by

$$g_P(x, y) = \left.\frac{\mathrm{d}V(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.12}$$

The empirical table averages the one-observation tables, and the first derivative is linear in the table change. Averaging the corresponding sensitivities therefore gives a first-order approximation to the error in $\widehat{V}(P)$. For independent observations, Section 4.3 will show that the variance of this average gives

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\mathrm{Var}_P\{g_P(X,Y)\}}{n_P}.$$

To evaluate the right-hand side, we first need an explicit expression for $g_P(x,y)$.

Differentiating Equation (4.10) separates this expression into the changes in the second moment and squared MI:

$$g_P(x,y) = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\Big|_0 - 2I(P)\,\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\Big|_0. \tag{4.13}$$

Both terms depend on how pointwise MI changes, which in turn depends on the joint and marginal probabilities. Along the chosen path, an arbitrary cell (i,j) has probability

$$p_{ij}(\varepsilon) = (1-\varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i=x, j=y\}. \tag{4.14}$$

The joint and marginal probability derivatives at the original distribution are

$$\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\Big|_0 = \mathbf{1}\{i=x, j=y\} - p_{ij}, \tag{4.15}$$

$$\frac{\mathrm{d}p_{i+}(\varepsilon)}{\mathrm{d}\varepsilon}\Big|_0 = \mathbf{1}\{i=x\} - p_{i+}, \tag{4.16}$$

$$\frac{\mathrm{d}p_{+j}(\varepsilon)}{\mathrm{d}\varepsilon}\Big|_0 = \mathbf{1}\{j=y\} - p_{+j}. \tag{4.17}$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j}\{\mathbf{1}(i=x, j=y) - p_{ij}\} = 1 - 1 = 0. \tag{4.18}$$

Effect of the perturbation on pointwise MI and MI.

The joint and marginal derivatives now determine the change in pointwise MI. Along the perturbation path,

$$\ell_{P_\varepsilon}(i,j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \tag{4.19}$$

For each logarithm, the derivative is the probability derivative divided by the probability. Substituting the joint and marginal derivatives gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\ell_{P_\varepsilon}(i,j)\Big|_0 &= \frac{\mathbf{1}\{i=x,j=y\}-p_{ij}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}-p_{i+}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}-p_{+j}}{p_{+j}} \\ &= \frac{\mathbf{1}\{i=x,j=y\}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}}{p_{+j}} + 1. \end{aligned} \tag{4.20}$$

The joint term supplies -1, while subtracting the row and column terms supplies $+1$ each. Their sum is the final $+1$.

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j). \tag{4.21}$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned}\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\Big|_0 = &\sum_{i,j}\{\mathbf{1}(i=x,j=y)-p_{ij}\}\ell_P(i,j) \\ &+ \sum_{i,j} p_{ij}\,\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\Big|_0 . \end{aligned} \tag{4.22}$$

The first sum is $\ell_P(x,y) - I(P)$. Evaluating the second sum using Equation (4.20), the selected-cell term contributes $p_{xy}/p_{xy} = 1$; the row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes -1; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\Big|_0 = \ell_P(x,y) - I(P)}. \tag{4.23}$$

This is the centred pointwise-MI influence function from Section 2.3.

Assembling the derivative of the pointwise-MI variance.

The MI derivative supplies the second part of Equation (4.13). To obtain the first part, it remains to differentiate the second moment,

$$\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j)^2. \tag{4.24}$$

Applying the product rule to the probability weights and squared pointwise-MI values gives

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\Big|_0 = \sum_{i,j}\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\Big|_0\,\ell_P(i,j)^2 + 2\sum_{i,j}p_{ij}\ell_P(i,j)\,\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\Big|_0. \tag{4.25}$$

The change in the probability weights contributes

$$\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\}. \tag{4.26}$$

To express the margin changes in the second sum, define the probability-weighted row and column means

$$\mathrm{E}_P\{\ell_P(X,Y)\mid X=x\} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}}, \tag{4.27}$$

$$\mathrm{E}_P\{\ell_P(X,Y)\mid Y=y\} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}}. \tag{4.28}$$

To evaluate the sum multiplied by 2 in Equation (4.25), substitute the pointwise-MI derivative from Equation (4.20). The sum has four terms. The indicators in the first three select the chosen cell, its row, and its column:

$$\sum_{i,j}p_{ij}\ell_P(i,j)\frac{\mathbf{1}\{i=x,j=y\}}{p_{ij}} = \ell_P(x,y),$$

$$\sum_{i,j}p_{ij}\ell_P(i,j)\frac{\mathbf{1}\{i=x\}}{p_{i+}} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}},$$

$$\sum_{i,j}p_{ij}\ell_P(i,j)\frac{\mathbf{1}\{j=y\}}{p_{+j}} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}},$$

$$\sum_{i,j}p_{ij}\ell_P(i,j) = I(P).$$

The row indicator fixes $i = x$ while leaving the sum over columns, so dividing by p_{x+} gives the conditional mean within row x. The column term works in the same way. The final term comes from the constant $+1$ in the pointwise-MI derivative and averages over the entire table. Keeping the minus signs on the row and column terms gives

$$\begin{aligned}
&\sum_{i,j} p_{ij}\ell_P(i,j)\,\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\bigg|_0 \\
&\quad = \ell_P(x,y) - \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}} - \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}} + I(P) \\
&\quad = \ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} - \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P). \qquad (4.29)
\end{aligned}$$

Combining Equations (4.23), (4.13), (4.26), and (4.29) gives the variance sensitivity. To see how its squared term arises, combine the weight-change term in Equation (4.26) with the derivative of $-I(P)^2$:

$$\begin{aligned}
&\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\} - 2I(P)\{\ell_P(x,y) - I(P)\} \\
&\quad = \ell_P(x,y)^2 - \{V(P) + I(P)^2\} - 2I(P)\ell_P(x,y) + 2I(P)^2 \\
&\quad = \ell_P(x,y)^2 - 2I(P)\ell_P(x,y) + I(P)^2 - V(P) \\
&\quad = \{\ell_P(x,y) - I(P)\}^2 - V(P).
\end{aligned}$$

The second line uses $\mathrm{E}_P\{\ell_P(X,Y)^2\} = V(P) + I(P)^2$; the last line completes the square. Adding twice the expression in Equation (4.29) gives

$$\boxed{\begin{aligned}
g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P) \\
+ 2\big[\ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} \\
- \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)\big].
\end{aligned}} \qquad (4.30)$$

The first line describes the change with the pointwise-MI values held fixed, while the bracket accounts for changes in those values as the cell probabilities and their row and column totals change. The conditional means collect the effects of the changing totals, so both parts are needed to describe how the estimated variance varies across samples.

4.3 Component degrees of freedom and final test

From individual observations to the error in the variance estimate.

We now express $\widehat{V}(P) - V(P)$ as an average of the sensitivities derived in the preceding section. The sampling mean and variance of this average will supply the two inputs to the component degrees of freedom.

Let $Z_a^{(P)}$ denote observation a from population P. It can be represented by a table that places probability one in its cell, denoted $\delta_{Z_a^{(P)}}$. Relative to P, that observation contributes the table change $\delta_{Z_a^{(P)}} - P$. The empirical table averages these one-observation tables, so its difference from P is

$$\widehat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \left\{ \delta_{Z_a^{(P)}} - P \right\}. \tag{4.31}$$

The plug-in estimate $\widehat{V}(P)$ evaluates V at the empirical table, so $\widehat{V}(P) = V(\widehat{P})$. Near a positive population table, V is smooth, so we approximate its change by a first-order Taylor expansion. For a differentiable scalar function expanded at zero, the general formula is

$$f(t) \approx f(0) + t f'(0).$$

Here set $f(t) = V\{P + t(\widehat{P} - P)\}$. As the scalar t increases, the table moves along the straight line from the population table P at $t = 0$ to the empirical table $\widehat{P}$ at $t = 1$. Thus V is expanded at P, the new table is $\widehat{P}$, and the table difference is $\widehat{P} - P$. The size of this change is determined by the cell-probability errors $\widehat{P} - P$, while t specifies a position along the path. The approximation requires these probability errors to be small. Since $f(0) = V(P)$ and $f(1) = \widehat{V}(P)$, evaluating the Taylor formula at $t = 1$ and subtracting $f(0)$ gives

$$\widehat{V}(P) - V(P) \approx f'(0) = \frac{\mathrm{d}}{\mathrm{d}t} V\{P + t(\widehat{P} - P)\}\Big|_{t=0}.$$

The omitted terms are quadratic or higher order in the cell-probability errors, so the approximation can be useful at $t = 1$ when these errors are small.

At fixed P, the linear Taylor term is a weighted sum of the cell-probability changes, with coefficients evaluated at P. These coefficients stay fixed when the direction of perturbation changes, so the derivative of an average table change is the average of the individual derivatives (van der Vaart, 1998, Chapter 20).

Substituting the empirical-table identity above therefore gives

$$\frac{\mathrm{d}}{\mathrm{d}t}V\{P+t(\widehat{P}-P)\}\Big|_{t=0} = \frac{1}{n_P}\sum_{a=1}^{n_P}\frac{\mathrm{d}}{\mathrm{d}t}V\{P+t(\delta_{Z_a^{(P)}}-P)\}\Big|_{t=0}.$$

Each term follows the one-observation perturbation $P+t(\delta_{Z_a^{(P)}}-P) = (1-t)P+t\delta_{Z_a^{(P)}}$, so its derivative is $g_P(Z_a^{(P)})$ by Equation (4.12). Combining this derivative identity with the Taylor approximation gives

$$\widehat{V}(P) - V(P) \approx \frac{1}{n_P}\sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \tag{4.32}$$

The variance-estimation error now has the same sample-average form as the MI error in Equation (2.25), with g_P supplying the observation-level contribution.

Sampling mean and variance of the estimate.

The linearised variance-estimation error averages the sensitivities g_P, so its mean is zero if their mean is zero. We verify this property by averaging the two parts of Equation (4.30). The first part has mean

$$\mathrm{E}_P\left[\{\ell_P(X,Y) - I(P)\}^2 - V(P)\right] = V(P) - V(P) = 0. \tag{4.33}$$

For the bracket, the row and column conditional means are averaged by weighting each row or column by its probability:

$$\begin{aligned}\mathrm{E}_P\left[\mathrm{E}_P\{\ell_P(X,Y) \mid X\}\right] &= \sum_x p_{x+}\frac{\sum_y p_{xy}\ell_P(x,y)}{p_{x+}} \\ &= \sum_{x,y} p_{xy}\ell_P(x,y) = I(P), \end{aligned}\tag{4.34}$$

$$\begin{aligned}\mathrm{E}_P\left[\mathrm{E}_P\{\ell_P(X,Y) \mid Y\}\right] &= \sum_y p_{+y}\frac{\sum_x p_{xy}\ell_P(x,y)}{p_{+y}} \\ &= \sum_{x,y} p_{xy}\ell_P(x,y) = I(P). \end{aligned}\tag{4.35}$$

Each marginal probability cancels the denominator of its conditional mean, leaving the probability-weighted sum over all cells. The bracket therefore also has mean zero, and the full sensitivity satisfies

$$\mathrm{E}_P\{g_P(X,Y)\} = 0 + 2\{I(P) - I(P) - I(P) + I(P)\} = 0. \tag{4.36}$$

Taking expectations in Equation (4.32) and using the zero mean of g_P gives

$$\begin{aligned}\mathrm{E}\{\widehat{V}(P)\} - V(P) &\approx \frac{1}{n_P}\sum_{a=1}^{n_P}\mathrm{E}_P\{g_P(Z_a^{(P)})\}\\ &= 0.\end{aligned}$$

This establishes the first-order mean approximation used in Section 4.1. The other required moment depends on the variation of g_P across possible observations. Define this population variance by

$$\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}. \tag{4.37}$$

Since g_P has mean zero, $\tau^2(P)$ measures variation in the sensitivities around zero. The independent observations give independent sensitivity contributions, so their covariance terms vanish in the following first-order sampling-variance approximation:

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{1}{n_P^2}\sum_{a=1}^{n_P}\mathrm{Var}_P\{g_P(Z_a^{(P)})\} \tag{4.38}$$

$$= \frac{\tau^2(P)}{n_P}. \tag{4.39}$$

Population component degrees of freedom.

The mean and variance approximations now supply both inputs to Equation (4.7). To obtain positive, finite component degrees of freedom, we require positive sensitivity variances:

ASSUMPTION 5 (Positive variances of observation-level sensitivities). *For the first-order construction of the Expanded Welch test,* $\tau^2(P) > 0$ *and* $\tau^2(Q) > 0$.

Substituting the first-order mean and variance approximations into Equation (4.7) gives the population first-order approximation to $\nu_{P,\mathrm{mom}}$, defined by

$$\boxed{\nu_{P,\mathrm{FO}} = \frac{2n_P V(P)^2}{\tau^2(P)}}. \tag{4.40}$$

To interpret Equation (4.40), divide the sampling variance of $\widehat{V}(P)$ by $V(P)^2$, the square of the population pointwise-MI variance:

$$\frac{\mathrm{Var}\{\widehat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \tag{4.41}$$

so the component degrees of freedom are approximately twice the reciprocal of this relative variance. When $\widehat{V}(P)$ fluctuates little relative to $V(P)$, the degrees of freedom are large and the Student t distribution is close to the standard normal distribution. Greater relative fluctuation instead gives fewer degrees of freedom and a larger rejection cutoff. Positive $V(P)$ does not guarantee positive $\tau^2(P)$: table changes can have no first-order effect on $V(P)$. In that case, describing the sampling variation of $\widehat{V}(P)$ requires a higher-order approximation.

Table-based component degrees of freedom.

The population inputs $V(P)$ and $\tau^2(P)$ are unknown, so the test estimates them from the observed table. We replace every population quantity in Equation (4.30) by its empirical counterpart and sum over occupied cells, since pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned}\widehat{g}_P(i,j) = \{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2 - \widehat{V}(P) \\ + 2\left[\widehat{\ell}_P(i,j) - \frac{\sum_{j'} \widehat{p}_{ij'}\widehat{\ell}_P(i,j')}{\widehat{p}_{i+}} - \frac{\sum_{i'} \widehat{p}_{i'j}\widehat{\ell}_P(i',j)}{\widehat{p}_{+j}} + \widehat{I}(P)\right].\end{aligned} \tag{4.42}$$

An empty cell has zero weight in these sums, so no logarithm or sensitivity needs to be evaluated there. Although the empirical sensitivities have probability-weighted mean zero under exact substitution, their variance is centred explicitly:

$$\widehat{\tau}^2(P) = \sum_{i,j} \widehat{p}_{ij}\left\{\widehat{g}_P(i,j) - \sum_{k,l}\widehat{p}_{kl}\widehat{g}_P(k,l)\right\}^2. \tag{4.43}$$

The estimated component degrees of freedom are

$$\boxed{\widehat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\widehat{\tau}^2(P)}}. \tag{4.44}$$

Repeating the same calculation for the second table gives $\widehat{\nu}_Q$.

Table 4.1 distinguishes the sampling-moment target, its population first-order approximation, and the sample-based estimate. The target and population approximation are fixed for a population and sample size, whereas $\widehat{\nu}_P$ changes with each sampled table.

TABLE 4.1: Component degrees of freedom and their inputs.

Quantity	Role	Inputs
$\nu_{P,\mathrm{mom}}$	Sampling-moment target	$\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{Var}\{\widehat{V}(P)\}$ at fixed P, n_P
$\nu_{P,\mathrm{FO}}$	Population first-order approximation to the target	Population $V(P)$, $\tau^2(P)$, and n_P
$\widehat{\nu}_P$	Component estimate used by Expanded Welch	$\widehat{V}(P)$, $\widehat{\tau}^2(P)$, and n_P from one sample

Combining the two samples' degrees of freedom.

The combining rule follows by applying moment matching to the full estimated variance,

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}.$$

Linearity of expectation gives its sampling mean:

$$\mathrm{E}(\widehat{\mathrm{SE}}^2) = \frac{\mathrm{E}\{\widehat{V}(P)\}}{n_P} + \frac{\mathrm{E}\{\widehat{V}(Q)\}}{n_Q}.$$

The two pointwise-MI variance estimates are computed from independent samples, so their covariance is zero and their contributions to $\mathrm{Var}(\widehat{\mathrm{SE}}^2)$ add. Rearranging Equation (4.7) expresses each component's sampling variance in terms of its sampling mean and moment-matched degrees of freedom. Applying this identity to both components gives

$$\begin{aligned}\mathrm{Var}(\widehat{\mathrm{SE}}^2) &= \frac{\mathrm{Var}\{\widehat{V}(P)\}}{n_P^2} + \frac{\mathrm{Var}\{\widehat{V}(Q)\}}{n_Q^2} \\ &= 2\left\{\frac{[\mathrm{E}\{\widehat{V}(P)\}/n_P]^2}{\nu_{P,\mathrm{mom}}} + \frac{[\mathrm{E}\{\widehat{V}(Q)\}/n_Q]^2}{\nu_{Q,\mathrm{mom}}}\right\}.\end{aligned}$$

The combined moment-matched degrees of freedom are twice the squared mean of $\widehat{\mathrm{SE}}^2$ divided by its variance. Substituting the expressions above and cancelling the common factor 2 gives

$$\frac{2[\mathrm{E}(\widehat{\mathrm{SE}}^2)]^2}{\mathrm{Var}(\widehat{\mathrm{SE}}^2)} = \frac{\left\{\mathrm{E}\{\widehat{V}(P)\}/n_P + \mathrm{E}\{\widehat{V}(Q)\}/n_Q\right\}^2}{[\mathrm{E}\{\widehat{V}(P)\}/n_P]^2/\nu_{P,\mathrm{mom}} + [\mathrm{E}\{\widehat{V}(Q)\}/n_Q]^2/\nu_{Q,\mathrm{mom}}}.$$

This establishes the combining rule in Equation (4.8) for population moment inputs. For the observed tables, we replace the sampling means $\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{E}\{\widehat{V}(Q)\}$ by $\widehat{V}(P)$ and $\widehat{V}(Q)$, and the component targets $\nu_{P,\mathrm{mom}}$ and $\nu_{Q,\mathrm{mom}}$ by $\widehat{\nu}_P$ and $\widehat{\nu}_Q$. These substitutions give the estimated combined degrees of

freedom:

$$\boxed{\widehat{\nu}_{\text{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\widehat{\nu}_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\widehat{\nu}_Q}}. \tag{4.45}$$

The two-sided p-value for the Expanded Welch test is

$$p_{\text{expanded}} = 2\left\{1 - F_t(|T|; \widehat{\nu}_{\text{expanded}})\right\}, \tag{4.46}$$

where $F_t(x;\nu)$ denotes the cumulative distribution function of a Student t distribution with ν degrees of freedom, evaluated at x.

For a fixed finite T, the two-sided p-value from the Student t distribution is at least as large as the Wald p-value from the standard normal distribution. We call a test calculation successful when all required quantities satisfy the numerical conditions in Appendix B.3. Every successful calculation for the Expanded Welch test also satisfies the Wald test's calculation requirements. At the same p-value threshold, every rejection by the Expanded Welch test is therefore also a rejection by the Wald test, including when unsuccessful calculations count as non-rejections in the reported rates. At the stated significance level, the Expanded Welch test cannot raise rejection where the Wald test is already conservative.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \qquad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \tag{4.47}$$

Their totals are $n_P = 105$ and $n_Q = 100$. In N^P, cell $(1,1)$ has count 35, and its row and column each total 50. Its empirical pointwise MI is therefore

$$\widehat{\ell}_P(1,1) = \log\left\{\frac{35/105}{(50/105)(50/105)}\right\} = \log(1.47) \simeq 0.38526.$$

This cell contributes $(35/105)(0.38526)$ to the plug-in MI estimate. Summing the contributions from all nine cells gives $\widehat{I}(P) = 0.13693$; the same calculation for N^Q gives $\widehat{I}(Q) = 0.25848$ nats. Since

$(3-1)(3-1)=4$, the corrected difference is

$$\widehat{\Delta}_{\mathrm{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \tag{4.48}$$

The variance calculation weights each squared deviation from the estimated MI by its cell probability. Cell $(1,1)$, for example, contributes $(35/105)(0.38526-0.13693)^2 \simeq 0.02056$ to $\widehat{V}(P)$. Summing over all cells gives $\widehat{V}(P)=0.27189$ and $\widehat{V}(Q)=0.43371$. Hence

$$\widehat{\mathrm{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \qquad T = -1.44898. \tag{4.49}$$

The Wald test gives $p=0.14734$.

For the Expanded Welch test, the same cell illustrates the sensitivity calculation. The first row's pointwise-MI values are 0.38526, -0.51083, and -0.64436, weighted by its counts 35, 10, and 5. Its conditional mean is

$$\frac{35(0.38526)+10(-0.51083)+5(-0.64436)}{50} \simeq 0.10308.$$

Because this count table is symmetric, the first column has the same conditional mean. Substituting these values into Equation (4.42) gives

$$\begin{aligned}\widehat{g}_P(1,1) &\simeq (0.38526-0.13693)^2 - 0.27189 \\ &\quad + 2\{0.38526 - 0.10308 - 0.10308 + 0.13693\} \\ &\simeq 0.4218.\end{aligned}$$

The probability-weighted mean of the sensitivities is zero, so this cell contributes $(35/105)(0.4218)^2 \simeq 0.0593$ to $\widehat{\tau}^2(P)$. Repeating the calculation for all cells in both tables gives $\widehat{\tau}^2(P)=0.93212$ and $\widehat{\tau}^2(Q)=0.82035$. Substitution into Equation (4.44) gives component degrees of freedom

$$\widehat{\nu}_P = 16.65, \qquad \widehat{\nu}_Q = 45.86. \tag{4.50}$$

Combining them with Equation (4.45) gives $\widehat{\nu}_{\mathrm{expanded}}=59.03$ and $p=0.15264$. Neither method rejects at the 5% level.

Operational algorithm and complexity.

Table 4.2 summarises one comparison.

TABLE 4.2: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their specified numbers of row and column categories; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\widehat{V}(P)$, $\widehat{V}(Q)$, the shared standard error, and T.
4	Compute the row and column pointwise-MI means needed in $\widehat{g}_P$ and $\widehat{g}_Q$.
5	Compute $\widehat{\tau}^2(P)$, $\widehat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.45).
7	Return the p-value from the Student t distribution, effective degrees of freedom, and results of the calculation checks.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$.

4.5 Behaviour and limitations

Near independence.

Consider a population table that approaches independence while its row and column margins stay fixed. At independence, every pointwise-MI value is zero. Since the linear MI error in Equation (2.22) weights each cell-probability error by its population pointwise MI, this linear contribution vanishes. Sampling errors can still create apparent association through products of cell-probability errors, which are retained by a second-order expansion. Near independence, the small linear weights allow these quadratic terms to contribute substantially to the MI estimation error.

The fixed-margin expansion in Appendix A.1 relates MI, the pointwise-MI variance $V(P)$, and the variance of the sensitivities $\tau^2(P)$ under the assumptions stated there. Close to independence along that path, with fixed table dimensions and positive cell probabilities, the expansion gives

$$V(P) \approx 2I(P), \qquad \tau^2(P) \approx 4V(P). \tag{4.51}$$

Substituting these local approximations into the component degrees-of-freedom formula gives

$$\nu_{P,\mathrm{FO}} = \frac{2n_P V(P)^2}{\tau^2(P)} \approx \frac{n_P V(P)}{2} \approx n_P I(P). \tag{4.52}$$

At a fixed sample size, this population first-order value tends to zero as positive population MI approaches zero. When $\widehat{V}(P)$ and $\widehat{\tau}^2(P)$ estimate their population counterparts accurately enough for the component formula to be reliable, $\widehat{\nu}_P$ follows this prediction: smaller MI gives fewer degrees of freedom and a larger cutoff from the Student t distribution.

The empirical table, however, need not be independent even when the population is independent. Random differences between its observed cell proportions and the products of its observed margins produce nonzero pointwise-MI values in occupied cells. Both $\widehat{V}(P)$ and $\widehat{\tau}^2(P)$ are calculated from these values, so sampling-induced association enters the estimated degrees of freedom. At a fixed sample size, these estimated inputs need not track the shrinking population quantities accurately enough for $\widehat{\nu}_P$ to follow $n_P I(P)$. Consequently, the estimated degrees of freedom need not approach zero with population MI. Appendix A.2 retains the quadratic sampling terms and derives their contribution to the estimated degrees of freedom.

Theoretical status and calculation requirements.

Equations (4.23) and (4.30) are exact derivatives for positive population tables. Using these derivatives to describe finite sampling errors gives first-order approximations for the MI estimate and the estimated pointwise-MI variance. Moment matching then approximates each variance contribution by a variable with a scaled chi-squared distribution, as in Equation (4.2). The exact construction for a Student t distribution in Section 4.1 also requires a numerator with a standard normal distribution and a squared denominator with a scaled chi-squared distribution. The numerator and squared denominator must be independent. Here, the corrected MI difference and its estimated standard error come from the same tables, so an observation can change both quantities. Matching the variance moments does not establish the required distributional or independence conditions, and the Student t distribution for T therefore remains a finite-sample approximation.

For fixed population tables satisfying the stated assumptions, including positive V and τ^2, the estimated degrees of freedom grow as the sample sizes tend to infinity, bringing the Student t distribution closer to the standard normal distribution used by the Wald test. This large-sample behaviour holds the populations fixed, whereas the near-independence comparison above reduces population MI at a fixed sample size.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error, while the Expanded Welch test also needs positive, finite component and combined degrees of freedom. If a requirement is not met, the calculation is unsuccessful and cannot be used to draw a test conclusion. The experiment applies the numerical calculation requirements in Appendix B.3 and assesses whether false-positive rates are close to 5%.

CHAPTER 5

Experimental Design

The simulations draw count tables from populations with known MI, allowing us to measure errors in the estimated MI difference and its standard error, and to count false positives when the population MI values are equal. The evaluation addresses the three research questions in Section 1.2 by checking the Expanded Welch test's variance approximations and comparing its false-positive rates and power with those of the Wald test. The design follows the simulation framework of Morris et al. (2019), specifying the populations, quantities being estimated, methods, and performance measures.

One observation records an (X, Y) outcome. One simulation repetition draws a sample of observations from each population, producing two count tables to which both tests are applied.

5.1 Setup: Constructing joint distributions with known MI

Before drawing samples, we construct population probability tables with specified MI values, using equal values to measure false-positive rates and different values to measure power. For each table, we first choose how frequently the categories occur, then change the association between X and Y while keeping those frequencies fixed.

The main simulations use square tables in which both variables have k categories, labelled $1, \ldots, k$, giving $k \times k$ cells in the joint distribution. Let $a = (a_1, \ldots, a_k)$ specify the category frequencies: a_i is the probability of category i, with $a_i > 0$ and $\sum_i a_i = 1$. Within one population, we use the same vector for both variables, so a specifies the row margins for X and the column margins for Y.

To obtain independent variables, we could draw X using a, then draw Y independently using a. Cell (i, j) would have probability $a_i a_j$, and MI would be zero. We introduce dependence by allowing Y to copy the category drawn for X with probability λ. For each observation, first draw X using a, then choose between two ways of generating Y, independently of the value of X:

- With probability λ, set $Y = X$.
- With probability $1 - \lambda$, draw Y independently using a.

Because copying guarantees a match whereas an independent draw matches X only by chance, increasing λ makes matching categories occur more often.

Both choices preserve the category frequencies of Y: copying inherits the probabilities a used to draw X, while an independent draw of Y uses a directly. Consequently, changing how often we copy changes which categories occur together while preserving the population row and column margins. Sample row and column proportions nevertheless fluctuate around these fixed probabilities.

Each cell probability is the sum of the contributions from these two choices. The independent choice, made with probability $1 - \lambda$, produces $X = i, Y = j$ with probability $a_i a_j$ and therefore contributes $(1 - \lambda)a_i a_j$. The copying choice contributes only to diagonal cells, where the categories match. Reaching cell (i, i) requires both copying and $X = i$, so its contribution is λa_i. Write $\mathbf{1}_{\{i=j\}}$ for an indicator that equals one when $i = j$ and zero otherwise. Adding the two contributions gives

$$p_{ij}(\lambda) = (1 - \lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}, \qquad 0 \leq \lambda < 1, \tag{5.1}$$

where the restriction $\lambda < 1$ keeps the independent choice possible and therefore gives every cell positive probability. This copying construction is a special case of the mixture model described by Bocci et al. (2010).

MI is zero at $\lambda = 0$ and increases with λ, although the copying probability needed to reach a particular MI depends on the category frequencies a. To construct a table with the desired MI, we hold a fixed and search numerically for λ, calculating MI from the population table at each candidate value. Appendix B gives the available MI range, the numerical search, and the fixed-margin calculation.

Constructing a 2×2 joint distribution with positive MI.

For population P, take $a = (0.7, 0.3)$, so each variable has probability 0.7 of taking category 1 and probability 0.3 of taking category 2. With a target MI of 0.02 nats, the numerical search gives a copying probability of approximately 0.20347.

Cell $(1, 1)$ requires $X = 1$, which has probability 0.7. Given $X = 1$, copying always gives $Y = 1$, while an independent draw gives $Y = 1$ with probability 0.7. Cell $(1, 2)$ also requires $X = 1$, but copying

cannot give $Y = 2$. It therefore requires the independent choice followed by a draw of category 2, which has probability 0.3. Multiplying these conditional probabilities by the probability of $X = 1$ gives

$$p_{11} \simeq 0.7\{0.20347 + (1 - 0.20347)(0.7)\} = 0.53273,$$
$$p_{12} \simeq 0.7(1 - 0.20347)(0.3) = 0.16727.$$

Both off-diagonal cells have probability $(1 - \lambda)(0.7)(0.3)$, so $p_{21} = p_{12}$. The second row totals 0.3, giving $p_{22} \simeq 0.3 - 0.16727 = 0.13273$.

For population Q, using $a = (0.8, 0.2)$ and a target MI of 0.03 nats gives a copying probability of approximately 0.26091. The resulting tables are

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \qquad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \tag{5.2}$$

These tables have $I(P) = 0.02$ and $I(Q) = 0.03$ nats, so the true signed difference is $\Delta = I(P) - I(Q) = -0.01$ nats. To construct an equal-MI comparison, we instead choose the same target MI for both populations and search for each copying probability separately.

Table sizes and margin regimes.

The experiment repeats this construction for square tables of sizes $2 \times 2, 4 \times 4, 8 \times 8$, and 16×16, containing 4, 16, 64, and 256 cells. Each doubling of the number of categories multiplies the number of cells by four, so a fixed sample must be distributed over more possible cells.

At each table size, we compare three patterns of category frequencies, called margin regimes. Uniform margins assign probability $1/k$ to every category, whereas uneven margins assign probability ρ to the first category and share the remaining probability $1 - \rho$ equally among the other $k - 1$ categories. We write this vector as

$$m_k(\rho) = \left(\rho, \frac{1-\rho}{k-1}, \ldots, \frac{1-\rho}{k-1}\right). \tag{5.3}$$

For example, assigning probability 0.8 to the first of four categories gives

$$m_4(0.8) = \left(0.8, \frac{0.2}{3}, \frac{0.2}{3}, \frac{0.2}{3}\right).$$

In Table 5.1, the same-skew regime uses the same uneven frequencies in both populations, whereas the different-skew regime uses a first-category probability of 0.7 in P and 0.8 in Q. Thus the row and column

margins agree within each population, while the margins can differ between populations. Here, "skew" describes uneven category probabilities, with no numerical ordering assumed for the category labels.

TABLE 5.1: Row and column margins for each population.

Regime	P: **rows and columns**	Q: **rows and columns**
Uniform	$(1/k, \ldots, 1/k)$	$(1/k, \ldots, 1/k)$
Same skew	$m_k(0.8)$	$m_k(0.8)$
Different skew	$m_k(0.7)$	$m_k(0.8)$

Under equal MI, the uniform and same-skew regimes give identical joint probability tables, $P = Q$: both populations have the same margins and the same copying probability. The different-skew regime instead gives different tables with equal MI because the category frequencies differ. Comparing the regimes therefore measures false positives under both identical tables and distinct tables with equal MI.

5.2 RQ1: Checking the variance approximations

The component moment target depends on the sampling mean $\mathrm{E}\{\widehat{V}(P)\}$ and variance $\mathrm{Var}\{\widehat{V}(P)\}$, with analogous quantities for Q. To estimate these moments, we hold the population table P and sample size n_P fixed, then draw repeated count tables and calculate $\widehat{V}(P)$ from each. The average of these estimates measures their sampling mean, and their sample variance measures their variation between samples.

The observation-level sensitivity derived in Chapter 4 has population variance $\tau^2(P)$ and describes how the estimated pointwise-MI variance responds to changes in the table. The first-order predictions are

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P), \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P}. \tag{5.4}$$

We compare $V(P)$ with the observed mean of $\widehat{V}(P)$, and $\tau^2(P)/n_P$ with its observed sampling variance. To examine the effect of estimating the sensitivity from each sample, we also average the sample-based prediction $\widehat{\tau}^2(P)/n_P$ across the repeated tables and compare this average with the same observed sampling variance.

We next compare the component degrees of freedom with the sampling-moment target from Chapter 4:

$$\nu_{P,\mathrm{mom}} = \frac{2\{\mathrm{E}\,\widehat{V}(P)\}^2}{\mathrm{Var}\{\widehat{V}(P)\}}. \tag{5.5}$$

Using the repeated samples to estimate the mean and variance in this target, we compare it with $\nu_{P,\mathrm{FO}}$, calculated from the population probabilities, and the average of the sampled $\widehat{\nu}_P$ values. Averaging $\widehat{\nu}_P$ need not recover $\nu_{P,\mathrm{mom}}$ because each estimate is a nonlinear ratio of sample-based quantities, whereas the target uses the sampling mean and variance of $\widehat{V}(P)$. For the sampled estimates, the ratio $\mathrm{mean}(\widehat{\nu}_P)/\nu_{P,\mathrm{mom}}$ equals one when the average estimate matches the target; a ratio below one means that the estimated degrees of freedom are lower on average. We make the same checks for Q and report the fraction of repetitions retained whenever a summary uses only positive finite degrees of freedom. Appendix B.6 gives the definitions and algebra.

The approximation checks in Table 5.2 use two grids: the first varies sample size at shared MI 0.02 nats, while the second includes very small positive MI to examine behaviour near independence. In this latter range, the predicted sampling variance of $\widehat{V}(P)$ becomes small and quadratic sampling terms can matter. Both grids use all three margin regimes and equal sample sizes $n_P = n_Q = n$. Each row combines every listed table size, MI value, and sample size.

TABLE 5.2: Settings for the approximation checks, broad null comparison, and power study.

Check	**Parameter values**
Across sample sizes	$k \in \{2, 4, 8, 16\}$, $I(P) = I(Q) = 0.02$ nats. $n \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}$.
Near independence	$k = 4$, $n = 1{,}000$. $I(P) = I(Q)$ takes the values (nats) $0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05$.
False-positive rates	$k \in \{2, 4, 8, 16\}$, $I(P) = I(Q) \in \{0.005, 0.02, 0.30\}$ nats. The same ten sample sizes as the across-sample-size checks.
Power at three MI gaps	$k \in \{2, 4, 8, 16\}$, $I(P) = 0.02$ nats, $I(Q) \in \{0.025, 0.03, 0.04\}$ nats. The same ten sample sizes as the across-sample-size checks.

5.3 RQ2: False-positive rates under equal MI

To measure false positives, we draw samples from populations with equal MI, so every rejection is a false positive. The copying construction sets

$$I(P) = I(Q) \in \{0.005, 0.02, 0.30\} \text{ nats}, \qquad \Delta = 0. \tag{5.6}$$

These levels cover weaker association, the baseline MI used for the power study, and stronger association. For each shared MI, we use all four table sizes, three margin regimes, and ten sample sizes in Tables 5.2 and 5.1. The resulting 360 settings test whether the methods' behaviour changes across those four experimental parameters.

At each setting, we draw 100,000 independent pairs of count tables and apply both tests to the same pair using their usual two-sided p-value threshold of 0.05. The false-positive rate is the fraction of all repetitions that reject equal MI, with unsuccessful calculations counted as non-rejections. We report the calculation success rate separately so calculation failures can be distinguished from conservative decisions.

A rejection rate above 5% indicates excessive rejection, while a rate below 5% indicates conservatism. We assess calibration by closeness to 5%, rather than treating every reduction in rejection as an improvement.

5.4 RQ3: Power across MI differences

To measure power, we draw samples from populations with different MI, so rejection detects a genuine difference. The power study uses the same four table sizes, three margin regimes, and ten sample sizes as the broad null comparison in Section 5.3. We repeat that grid for three MI gaps by holding $I(P) = 0.02$ nats and changing $I(Q)$:

$$\begin{aligned} I(P) = 0.02, \qquad I(Q) \in \{0.025, 0.03, 0.04\}, \\ \Delta = I(P) - I(Q) \in \{-0.005, -0.01, -0.02\} \text{ nats}. \end{aligned} \tag{5.7}$$

Thus Q has greater MI in each alternative, with gap magnitudes $|\Delta| \in \{0.005, 0.01, 0.02\}$ nats. Each magnitude uses all 120 combinations of table size, margin regime, and sample size, giving 360 power settings. Within each curve, the two population tables and MI gap stay fixed as sample size increases; comparing curves changes Q's association while retaining its margins.

For each setting, we generate 100,000 independent pairs of count tables and apply both tests to each pair using their usual p-value threshold of 0.05. Power is the fraction of all repetitions that reject equal MI, with unsuccessful calculations counted as non-rejections and the calculation success rate reported separately. Because the tests are two-sided, power counts rejections with either a positive or a negative estimated MI difference. The sample sizes and margin profiles are specified in Tables 5.2 and 5.1.

The Expanded Welch test uses the same statistic as the Wald test but has a larger rejection cutoff, so it cannot have higher power at the same p-value threshold of 0.05. We measure the reduction in detection probability across the grid and interpret it alongside the false-positive rates in Section 5.3, because a higher rejection rate under an alternative can accompany excessive rejection under equal MI.

5.5 Shared evaluation: Sampling and uncertainty

At one evaluation setting, each repetition draws a count table from P and an independent count table from Q. Both tests use these same two tables, so their rejection decisions can be compared on the same sampled data. Each setting has 100,000 independent repetitions, which determine the precision of the reported rates, whereas each test uses only n_P and n_Q observations within a repetition. Each distinct setting uses separate random streams for the two populations. Repeated settings use the same repetitions wherever they appear, whereas different MI-gap settings use independent simulation streams.

All reported simulation comparisons use equal sample sizes, so the two leading MI bias corrections cancel from $\widehat{\Delta}_{\mathrm{BC}}$, leaving the raw plug-in MI difference. Residual biases can still differ between distinct population tables, shifting the estimated difference away from its true value even under equal population MI.

Table 5.3 defines the measures for each evaluation setting.

TABLE 5.3: Performance measures calculated from repeated samples.

Measure	**Calculation**
Rejection rate	$\dfrac{\text{number of rejections}}{\text{number of simulation repetitions}}$
Calculation success rate	$\dfrac{\text{number of successful calculations}}{\text{number of simulation repetitions}}$
Bias in the estimated MI difference	$\mathrm{mean}(\widehat{\Delta}_{\mathrm{BC}}) - \Delta$
Variance ratio	$\dfrac{\mathrm{mean}(\widehat{\mathrm{SE}}^2)}{\text{observed variance of } \widehat{\Delta}_{\mathrm{BC}}}$

Bias uses finite MI differences even when the test calculation is unsuccessful. The variance ratio uses repetitions with both finite $\widehat{\Delta}_{\mathrm{BC}}$ and finite $\widehat{\mathrm{SE}}^2$, with one indicating agreement. Ratios below one mean

that the squared standard error understates the observed sampling variance of the estimated MI difference on average, while ratios above one indicate overstatement. Appendix B.4 specifies the inclusion rules and complete calculations.

Rejection rates have pointwise Wilson 95% confidence intervals (Wilson, 1927), which describe uncertainty from the finite number of simulation repetitions at each specified population and sample size. Appendix B.5 gives the corresponding Monte Carlo standard error.

CHAPTER 6

Results

This chapter checks the approximations used to choose the Expanded Welch test's degrees of freedom, then compares its false-positive rates and power with those of the Wald test. The online-shopping application illustrates how to interpret the test outputs.

Each evaluation setting uses 100,000 independent repetitions, each drawing one count table from P and one independently from Q. The Wald test and the Expanded Welch test are both applied to the same two sampled tables in each repetition.

6.1 RQ1: Accuracy of variance approximations

For a fixed population table P and sample size n_P, each repetition produces an estimate $\widehat{V}(P)$ of the pointwise-MI variance $V(P)$. We first check predictions for its sampling mean, $\mathrm{E}\{\widehat{V}(P)\}$, and its variation between repetitions, $\mathrm{Var}\{\widehat{V}(P)\}$, because both moments enter the component degrees-of-freedom calculation. We then examine how errors in these moments affect the degrees of freedom, before checking the approximations as the populations approach independence. Table 3.1 in Chapter 3 distinguishes these quantities from the sampling variance of the MI difference. The figures focus on populations with equal MI but different margins, allowing approximation accuracy to be compared between distinct population tables.

6.1.1 Mean and sampling-variance predictions

Equation (5.4) predicts $\mathrm{E}\{\widehat{V}(P)\} \approx V(P)$ and $\mathrm{Var}\{\widehat{V}(P)\} \approx \tau^2(P)/n_P$, where $\tau^2(P)$ measures the variability of the observation-level sensitivities derived in Chapter 4. We compare each prediction with the corresponding moment observed across repetitions, and repeat the checks for Q.

Figure 6.1 shows the different-skew populations at shared MI 0.02 nats, with row and column margins $m_k(0.7)$ for P and $m_k(0.8)$ for Q. The top row divides the simulated mean of $\widehat{V}(P)$ by the population benchmark $V(P)$, and likewise for Q. The middle row divides $\tau^2(P)/n_P$ by the observed $\mathrm{Var}\{\widehat{V}(P)\}$. The bottom row uses the average $\widehat{\tau}^2(P)/n_P$ as its numerator and the same observed variance as its denominator. Both variance rows use the same vertical scale, so their differences show the effect of substituting sample proportions. A ratio of one indicates agreement. Shading in the first two rows shows approximate 95% Monte Carlo intervals from 100,000 repetitions. Their half-widths are below 1.3% of the plotted values, making the bands barely visible on this scale.

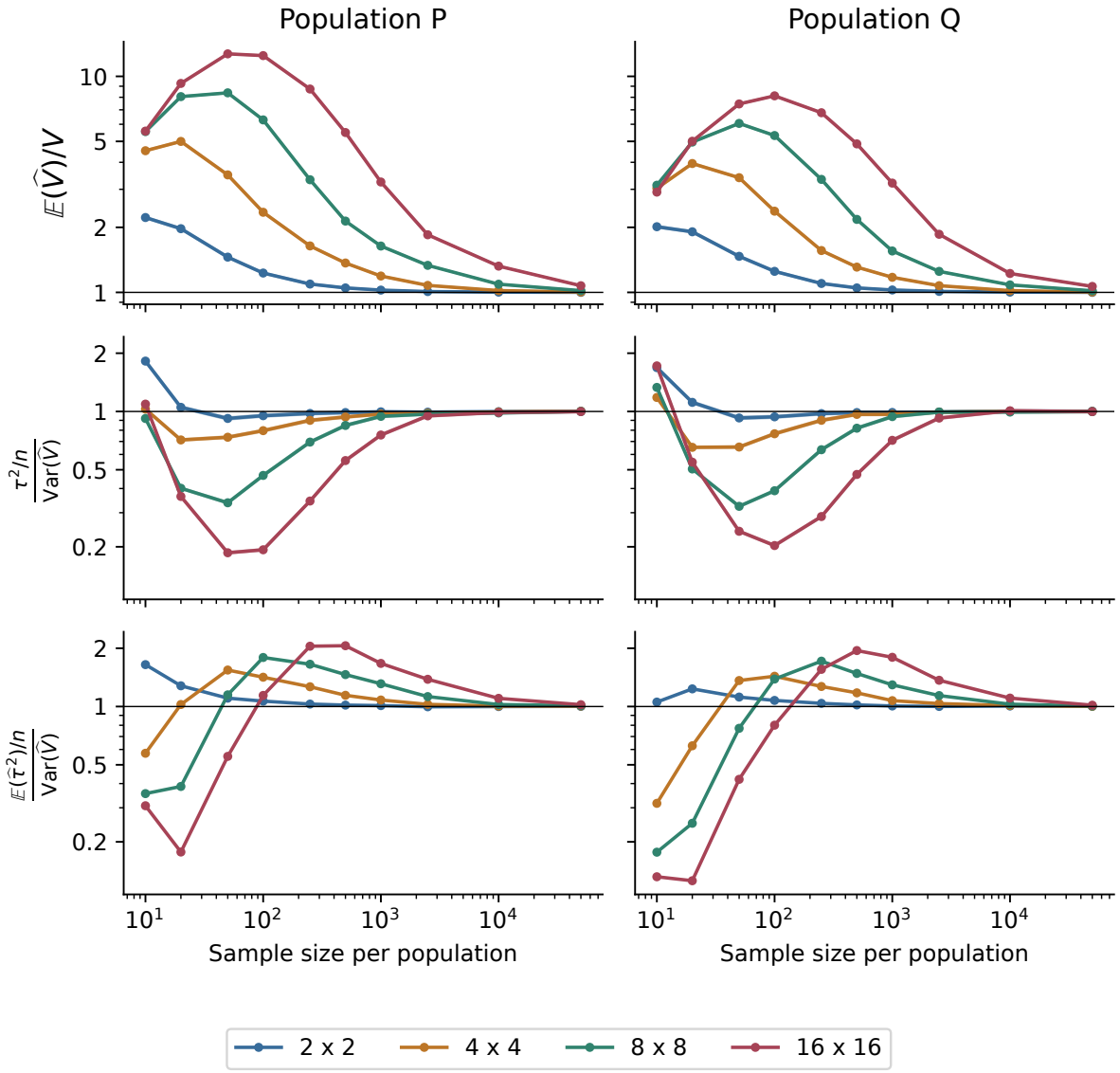


FIGURE 6.1: Mean and sampling-variance ratios under different-skew margins.

In the top row of Figure 6.1, larger tables have higher peaks in the mean ratio and take more observations to approach one. The 2×2 curve falls towards one without a pronounced peak, whereas the 4×4, 8×8, and 16×16 curves first rise and then fall. The 4×4 peak is smaller and earlier than those for the two larger tables. The rise in these curves occurs as rare combinations that were initially unobserved begin to appear in sampled tables. Cells observed only once can have estimated pointwise MI far from the table's plug-in MI, producing large squared deviations in $\widehat{V}$. For example, if a cell and its row and column each contain one observation, its empirical pointwise MI is $\log n$. A cell-contribution decomposition using 10,000 repetitions for the 16×16 populations shows that, at $n = 50$, once-observed cells account for about 88% of the mean of $\widehat{V}$ in P and 92% in Q (Appendix B.4). Their contribution falls as repeated counts give better estimates of the cell probabilities. Larger tables spread probability over more rare combinations under these margin rules, so this distortion persists to larger sample sizes. Both populations show these patterns, but the larger-table peaks are higher in P than in Q, and the 16×16 peak occurs later in Q. For each of these larger table sizes, the peak mean of $\widehat{V}$ is greater in P than in Q, while $V(P) < V(Q)$. Both the larger numerator and the smaller denominator therefore raise P's peak ratio. Equal MI fixes the population mean of pointwise MI, whereas its variance V also depends on the cell-specific pointwise-MI values and their probability weights, which change with the margins.

In the middle row, the 2×2 curves have shallow troughs, while the larger tables have progressively deeper troughs and take longer to approach one. Ratios below one mean that $\tau^2(P)/n_P$ understates the observed $\mathrm{Var}\{\widehat{V}(P)\}$, with the same interpretation for Q. The dip develops because, for a fixed population, the predicted variance decreases exactly as $1/n$, while the observed sampling variance does not initially follow that decrease. At small sample sizes, many rare combinations remain unobserved. As n increases, some sampled tables contain a given rare cell while others still miss it. An unobserved cell contributes zero to $\widehat{V}$, whereas a single observation can give a large squared deviation of its pointwise MI from the table's plug-in MI. The $1/n$ change in a cell proportion can be large relative to its true probability, so its effect on $\widehat{V}$ can be poorly represented by linearisation around the population table. The observed variance can therefore initially rise or fall more slowly than $1/n$, even though the predicted variance decreases as $1/n$. The ratio of predicted to observed variance therefore falls. To check how much variation the linear approximation misses, we separate $\widehat{V} - V$ into the predicted linear change and the remaining error. In the same 16×16 repetitions, this remaining error has greater variance than the linear change around the trough. After accounting for the covariance between the two parts, the observed variance still substantially exceeds the linear prediction (Appendix B.4, Equation (B.8)). With larger samples, repeated counts make the estimated cell probabilities and their log ratios more stable. The variation missed by the

linear approximation becomes smaller relative to $\tau^2(P)/n_P$, and the observed variance approaches this prediction, so the ratio rises back towards one. Under these margin rules, larger tables contain more rare combinations and require more observations before this approximation becomes accurate. The observed variance measures spread about the sampling mean, so it can be predicted accurately while that mean remains above the population benchmark, as in the top row. The same ordering by table size appears in P and Q, but the 16×16 trough occurs later in Q, where the rarer row and column categories have lower probabilities and accumulate observations more slowly.

In the bottom row, the 4×4, 8×8, and 16×16 sample-based predictions first underestimate the observed sampling variance, then overestimate it before approaching agreement. Their ratios begin below one, rise through one to a peak, and then fall towards one, although the 16×16 curves initially dip further before rising. The 4×4 overshoot is smaller and earlier than those for the two larger tables, while 16×16 has the largest overshoot and takes longest to settle near one. The 2×2 curves begin above one and settle near it sooner. Substituting sample proportions changes the sensitivities used to estimate τ^2. Around the overshoot, cells observed only once contribute strongly to $\widehat{\tau}^2$ as well as to $\widehat{V}$, so the averaged sample-based prediction can exceed the observed sampling variance even while the middle-row population prediction remains below it. Sample substitution can therefore reverse the direction of the prediction error. Both populations show this progression, but for the three larger table sizes the initial underprediction is stronger in Q, and the crossing above one occurs later than in P. The subsequent peaks are higher in P than in Q.

6.1.2 How moment errors affect degrees of freedom

We now examine how the moment errors in Figure 6.1 affect the component degrees of freedom for P and Q, using the three quantities and inputs distinguished in Table 4.1. At each setting, the observed mean and sampling variance of $\widehat{V}(P)$ across 100,000 repetitions give the moment target $\nu_{P,\mathrm{mom}}$ in Equation (5.5). The population first-order value $\nu_{P,\mathrm{FO}}$ instead uses $V(P)$ and $\tau^2(P)/n_P$, while the sample-based comparison calculates $\widehat{\nu}_P$ from each sampled table before averaging these estimates.

Figure 6.2 uses the different-skew populations at shared MI 0.02 nats, with margins $m_k(0.7)$ for P and $m_k(0.8)$ for Q. Sample size increases along the horizontal axis, and the lines represent table sizes. The top row divides $\nu_{P,\mathrm{FO}}$ by $\nu_{P,\mathrm{mom}}$; the bottom row divides $\mathrm{E}(\widehat{\nu}_P)$ by the same target, and likewise for Q. On the shared vertical scale, one indicates agreement, while values below or above one indicate fewer or

more degrees of freedom than the target. A bottom-row value is shown only when all 100,000 component estimates are positive and finite. Shading in the top row gives approximate 95% Monte Carlo intervals.

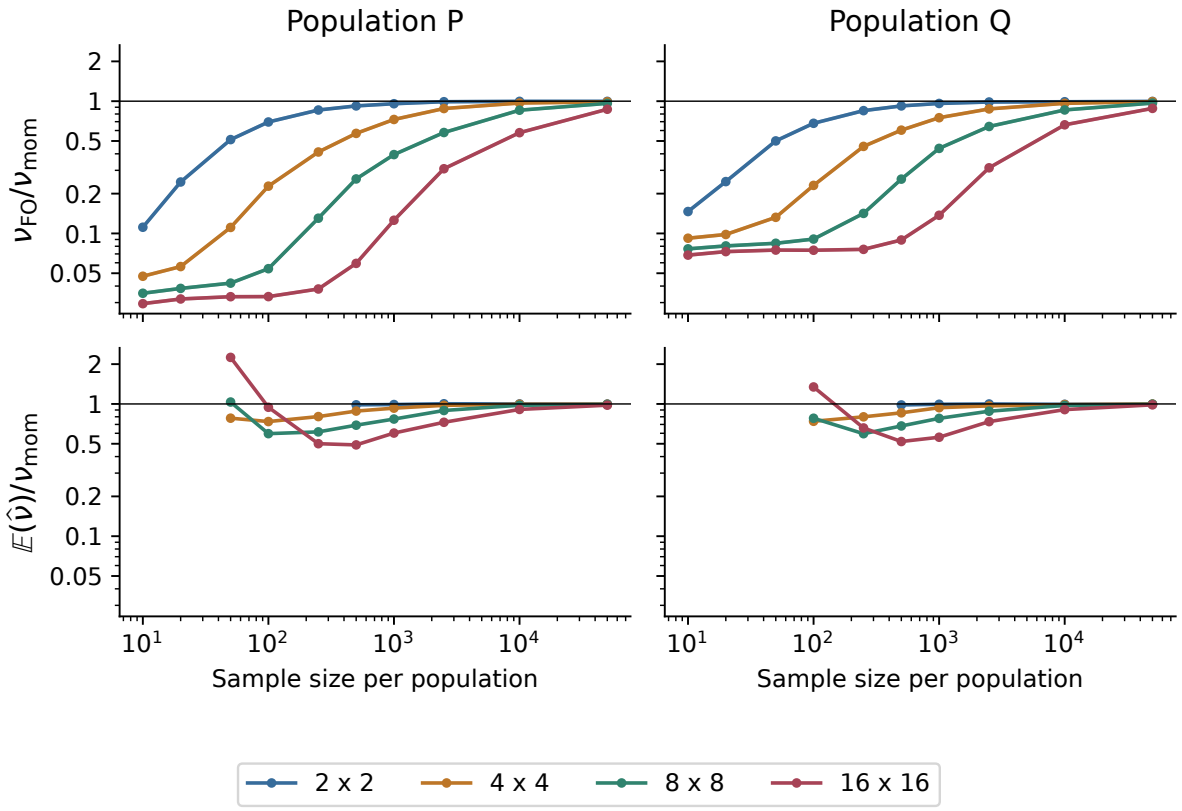


FIGURE 6.2: Component degrees-of-freedom approximations relative to their moment targets.

In the top row of Figure 6.2, the first-order values remain below the moment targets over most of the sample-size range. The 2×2 curves approach one first, followed by 4×4; the 8×8 and 16×16 curves remain lower for longer. This pattern arises because the two moment errors enter different parts of the degrees-of-freedom calculation. Replacing the enlarged sampling mean of $\widehat{V}(P)$ by the smaller $V(P)$ lowers the squared numerator. A variance prediction below the observed sampling variance lowers the denominator and raises the resulting degrees of freedom; a prediction above the observed variance lowers them. Over most of the range, the reduction from the squared mean outweighs the upward effect of an underestimated denominator. The larger tables have greater mean errors in Figure 6.1, so their first-order degrees of freedom remain farther below the target. As the moment approximations improve, both effects diminish and the ratios approach one. Although the same table-size ordering appears in P and Q, the larger-table ratios are generally closer to one in Q. In Q, the sampling mean of $\widehat{V}(Q)$ exceeds $V(Q)$ by a

smaller percentage than in P, so substituting $V(Q)$ for that mean causes a smaller proportional reduction in the squared numerator.

A numerical example shows why a fairly accurate variance prediction can still give degrees of freedom below the target. For different-skew 4×4 population P at $n = 1{,}000$ and MI 0.02, the sampling mean of $\widehat{V}(P)$ is about 19% above $V(P)$, whereas $\tau^2(P)/n_P$ is only about 3% below the observed sampling variance. The first-order calculation uses $V(P)$ in place of the larger sampling mean and squares their ratio, giving approximately

$$\frac{\nu_{P,\mathrm{FO}}}{\nu_{P,\mathrm{mom}}} \approx \frac{(1/1.19)^2}{0.97} \approx 0.73.$$

The squared mean ratio therefore dominates the modest variance error, consistent with first-order degrees of freedom of 15.30 compared with a moment target of 21.03.

In the bottom row, the 16×16 curves first appear above the target, fall below it, and then recover towards one. The 4×4 and 8×8 curves have smaller excursions, while the 2×2 curves are already close to one when they become visible. This change from the top row occurs because each sampled table supplies both the squared numerator $\widehat{V}(P)^2$ and the variance prediction $\widehat{\tau}^2(P)/n_P$. At the first shown 16×16 points, the variance prediction is too small on average, tending to raise the estimated degrees of freedom. As the prediction subsequently becomes too large, it tends to lower the estimated degrees of freedom before the variance prediction returns towards the observed sampling variance. The two inputs come from the same table, so their joint variation also affects the mean degrees of freedom when we average the sampled ratios rather than insert average inputs into a single ratio. With larger samples, both inputs become more stable and the mean estimated degrees of freedom approach the target. They can recover sooner than the population first-order values because their numerators use the sampled $\widehat{V}$, rather than the smaller fixed V. For the three larger tables, the curves appear later in Q because positive, finite component degrees of freedom in every repetition require more observations than in P. Missing earlier points indicate failed component calculations in some repetitions, rather than zero degrees of freedom.

6.1.3 Behaviour near independence

When X and Y are nearly independent within each population, the pointwise-MI weights in the linear sampling error become small, so products of cell-probability errors can contribute substantially to sampling variation. These quadratic terms allow sampled tables to show apparent association even when population MI is almost zero. Appendix A.2 retains these quadratic terms and approximates the scaled

MI estimate $2n\widehat{I}$ using a noncentral chi-squared distribution, which accounts for both sampling noise and a small population association. Together with the near-independence relation $\widehat{V} \approx 2\widehat{I}$, this approximation predicts the mean and sampling variance of $\widehat{V}$. Applying the local approximations to the component formula also predicts the mean estimated component degrees of freedom.

Figure 6.3 holds the sample size at $n = 1{,}000$ and the table size at 4×4, and increases the shared population MI along the horizontal axis. The margins are $m_k(0.7)$ for P and $m_k(0.8)$ for Q. At independence, the approximation predicts a sampling mean of $\widehat{V}$ proportional to $1/n$ and a sampling variance proportional to $1/n^2$. The top row therefore multiplies the observed sampling mean of $\widehat{V}$ by n, while the middle row multiplies its observed sampling variance by n^2, cancelling these sample-size factors. The dashed near-independence predictions use the same scalings and depend on the shared table size, sample size, and MI, so they are identical in the two columns. The bottom row compares the mean estimated degrees of freedom, its near-independence prediction, the sampling-moment target, and the population first-order value. Across the simulation repetitions, we calculate the observed means and variances and average the degrees of freedom estimated separately from each sampled table.

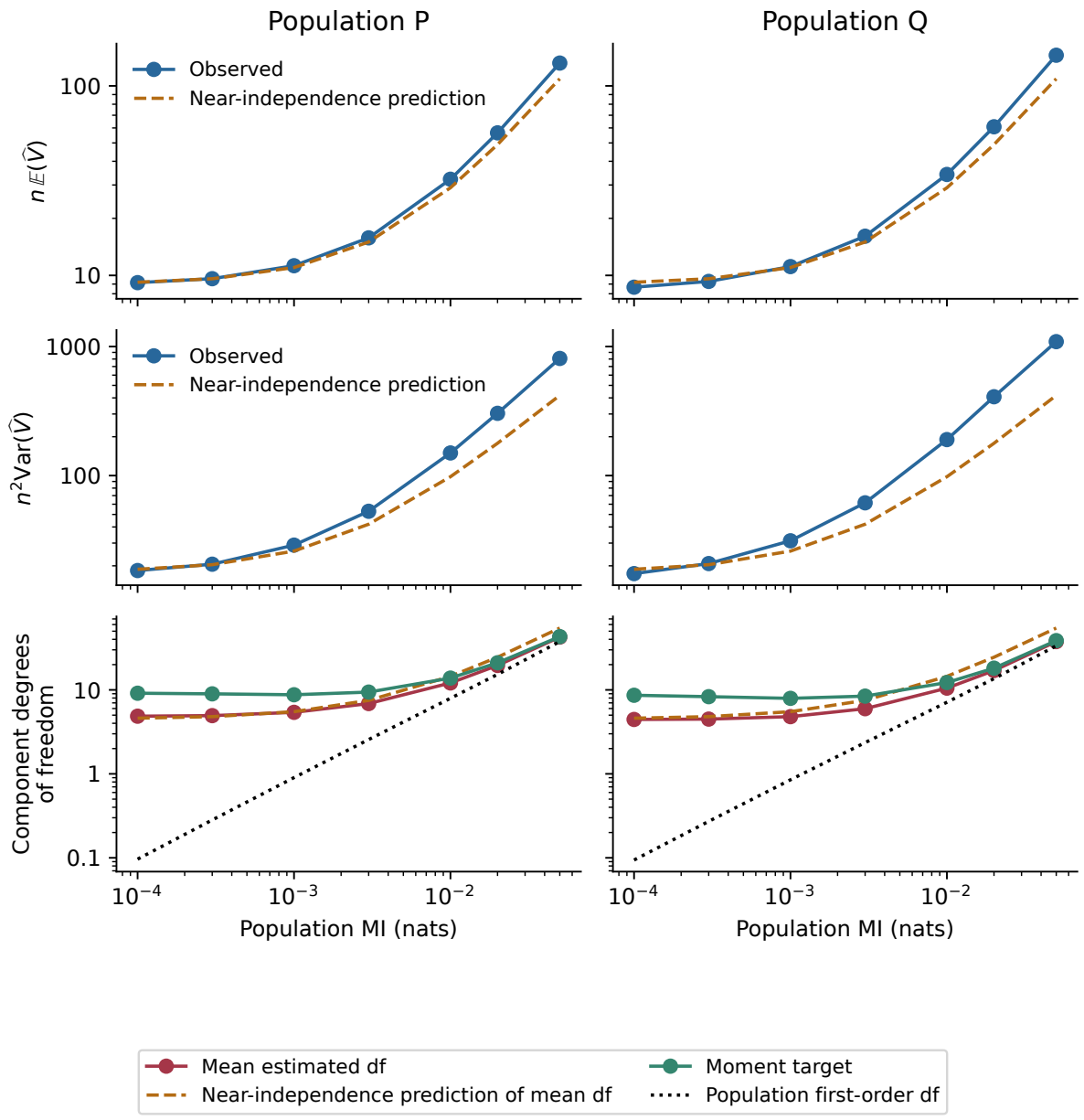


FIGURE 6.3: Component moments and degrees of freedom near independence.

In the top row of Figure 6.3, the observed scaled mean and its near-independence prediction nearly coincide at the smallest MI values, then both rise as association strengthens, with the observed mean eventually exceeding the prediction. The positive starting level occurs because sampling noise produces positive plug-in MI and $\widehat{V}$ even when population association is almost absent. As MI increases, the cell log ratios move farther from their values at independence, and the omitted terms in the local expansion become more important. Checks on the sampled tables show that $2\widehat{I}$ then increasingly understates $\widehat{V}$. The

observed means also separate between the columns, with Q higher at the larger MI values. The identical local predictions cannot capture this margin dependence because they retain only the shared table size, sample size, and MI.

In the middle row, the observed scaled sampling variance also starts close to the near-independence prediction, but their separation grows more strongly as MI increases. For the same 4×4, different-skew comparisons at $n = 1{,}000$, checks across increasing shared MI values trace this underprediction through the two steps of the local calculation. The calculation first represents $\widehat{V}$ by $2\widehat{I}$, then predicts the sampling variance of that quantity using the noncentral chi-squared distribution introduced above. At stronger association, $2\widehat{I}$ varies less between samples than $\widehat{V}$, while the local distribution also understates the observed variation in $2\widehat{I}$ itself. Both errors therefore lower the predicted sampling variance relative to the observed $\mathrm{Var}\{\widehat{V}\}$. The gap is larger than in the mean row in both columns, and the observed sampling variance is greater in Q than in P at the larger MI values. Thus the different margins affect sampling variation beyond the terms retained by the shared local prediction.

In the bottom row, the mean estimated degrees of freedom remain several units at the smallest MI values, around half the sampling-moment target, while the population first-order value approaches zero. This separation arises because the sample-based calculation retains apparent association from sampling noise. Substituting $\widehat{V} \approx 2\widehat{I}$ and $\widehat{\tau}^2 \approx 4\widehat{V}$ into the component degrees-of-freedom formula gives $\widehat{\nu}_P \approx n_P \widehat{I}(P)$, whose average stays positive at fixed sample size even when $I(P)$ is almost zero. The population first-order value instead approaches $n_P I(P)$, which becomes very small. The moment target has a shallow initial dip because the observed sampling variance of $\widehat{V}$ initially grows proportionally faster than the squared sampling mean of $\widehat{V}$. Their relative growth later reverses and the target rises, with the mean estimated degrees of freedom approaching it. At stronger association, the dashed near-independence prediction rises above the observed mean estimated degrees of freedom, while the population first-order curve remains below the moment target. The near-independence calculation approximates a ratio with $\widehat{V}$ squared in the numerator and $\widehat{\tau}^2$ in the denominator. In this range, replacing $\widehat{\tau}^2$ by $4\widehat{V}$ generally reduces the denominator and raises the mean estimated degrees of freedom. The subsequent substitution $\widehat{V} \approx 2\widehat{I}$ lowers that mean because $2\widehat{I}$ understates $\widehat{V}$, partly offsetting the first increase. The denominator effect is larger, so the local calculation overpredicts the mean estimated degrees of freedom even though it underpredicts the moments in the upper rows. Both populations show this progression, but the observed mean estimated degrees of freedom and the moment target are lower in Q.

The population first-order and average sample-based variance predictions err in opposite directions at the smallest MI values. We compare the predictions defined for the middle and bottom rows of Figure 6.1 using the same repetitions. At the smallest MI target, 0.0001, the population first-order predictions are only about 5% of the observed sampling variances, while the sample-based predictions average roughly twice those variances. The local relation $\widehat{\tau}^2 \approx 4\widehat{V}$ carries sampling noise into the estimated variance prediction, explaining the sample-based prediction's upward error in this range (Appendix A.2). At MI 0.02, the population first-order variance predictions are much more accurate, although the sample-based ratios remain above one. As MI increases, the variance of the linear sampling term better approximates the observed sampling variance of $\widehat{V}$, while the local near-independence calculation loses accuracy. At MI 0.05, the observed means of $\widehat{V}$ remain roughly 6% above V in both populations, while the population first-order and average sample-based variance predictions are close to the observed sampling variances. The mean estimated degrees of freedom are closer to the moment target than the population first-order values, which still use V in place of the enlarged sampling mean.

6.2 RQ2: False-positive rates under equal MI

The null study uses shared population MI values of 0.005, 0.02, and 0.30 nats, with all four table sizes, three margin regimes, and ten sample sizes at each level. Every population pair has $\Delta = 0$, so each rejection is a false positive. Each plotted rate divides the number of rejections by all 100,000 repetitions, with unsuccessful calculations counted as non-rejections. Shading gives pointwise 95% intervals, and the lower strips show the calculation success rates.

The two tests use the same statistic, but the Expanded Welch test has a larger rejection cutoff. At the same p-value threshold of 0.05, its rejection rate cannot exceed that of the Wald test. The comparison asks whether that reduction brings rejection closer to 5% or deepens conservatism. The three figures below follow the same margin order as the power study and show the effects of sample size, table size, and shared MI together.

6.2.1 Uniform margins

Figure 6.4 uses uniform margins in both populations, giving identical population tables at each shared MI. Each panel increases sample size along the logarithmic horizontal axis, with colour identifying shared MI. Solid lines with filled circles show the Wald test's false-positive rates, and dashed lines with open squares

show those of the Expanded Welch test. Where the method curves overlap, their observed false-positive rates are equal or too close to distinguish on the plotted scale. The dotted line marks 5%. For a given table size, the vertical scale is the same across the three margin figures, while different table sizes can use different scales.

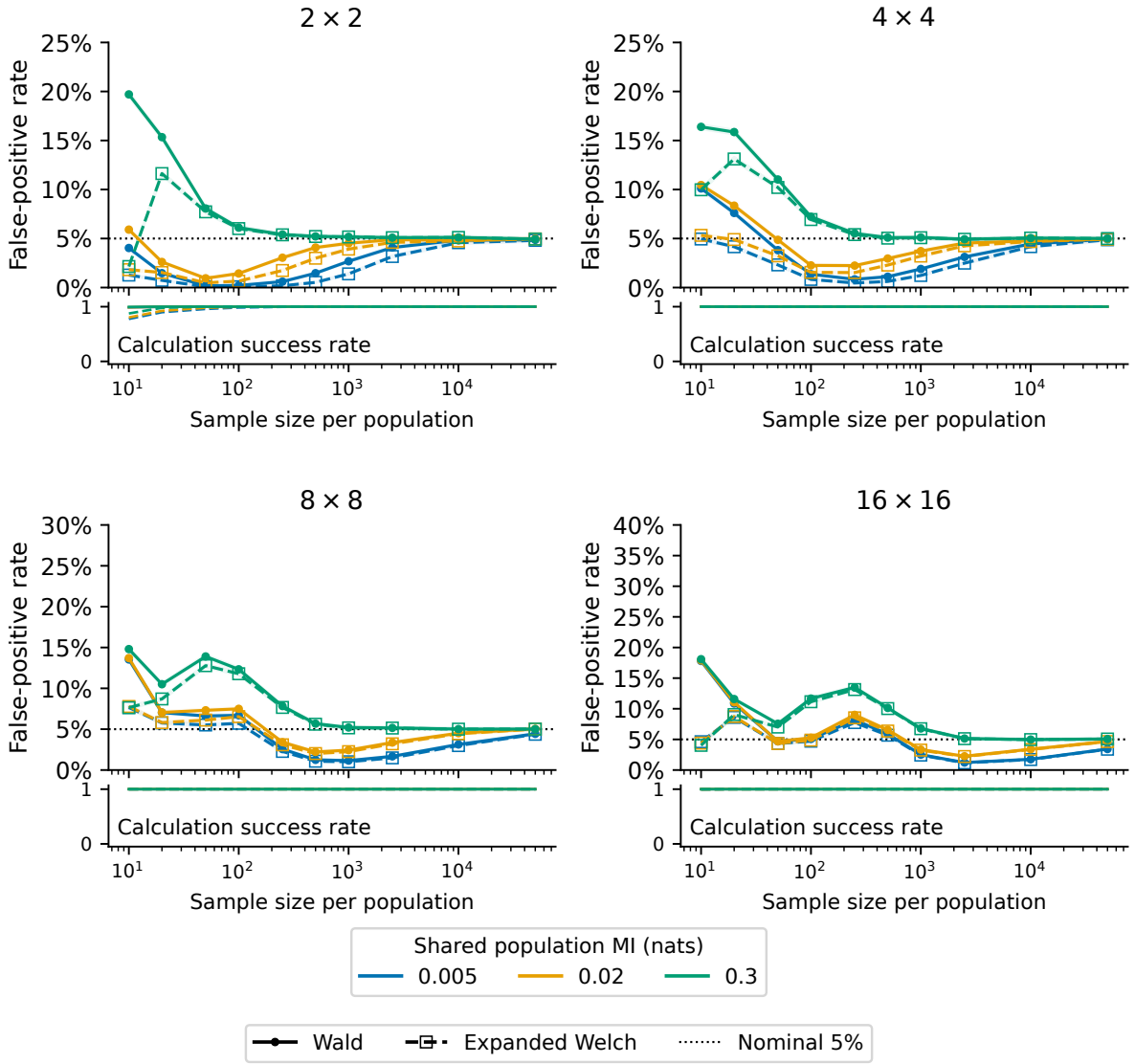


FIGURE 6.4: False-positive rates across shared MI values under uniform margins.

At shared MI 0.005 and 0.02, both tests have ranges of conservative rejection before recovering towards 5%, with later recovery for larger tables. At shared MI 0.30, the curves instead show substantial excessive rejection over parts of the smaller-sample range before settling near the target. Increasing shared MI therefore changes the rejection pattern, even though the population difference remains zero. The biases in

the two MI estimates cancel because the population tables and sample sizes are identical, so the departures from 5% arise through the estimated standard error and the sampling distribution of the shared statistic.

The Expanded Welch test lowers both the excessive and conservative parts of these curves. In 4×4 at $n = 20$, its false-positive rate of 4.90% is closer to 5% than the Wald test's rate of 8.35% at shared MI 0.02. At shared MI 0.30, however, the Expanded Welch test still rejects in 13.11% of repetitions at the same dimensions and sample size. The larger cutoff can correct some excess rejection without making the method uniformly accurate as shared MI changes. Conversely, where the Wald test's false-positive rate is already below 5%, the reduction moves rejection farther from the target.

6.2.2 Same-skew margins

Figure 6.5 assigns both populations the row and column profile $m_k(0.8)$, again giving identical tables under equal MI. This comparison retains the same MI levels and sample-size grid while concentrating more probability in the first category.

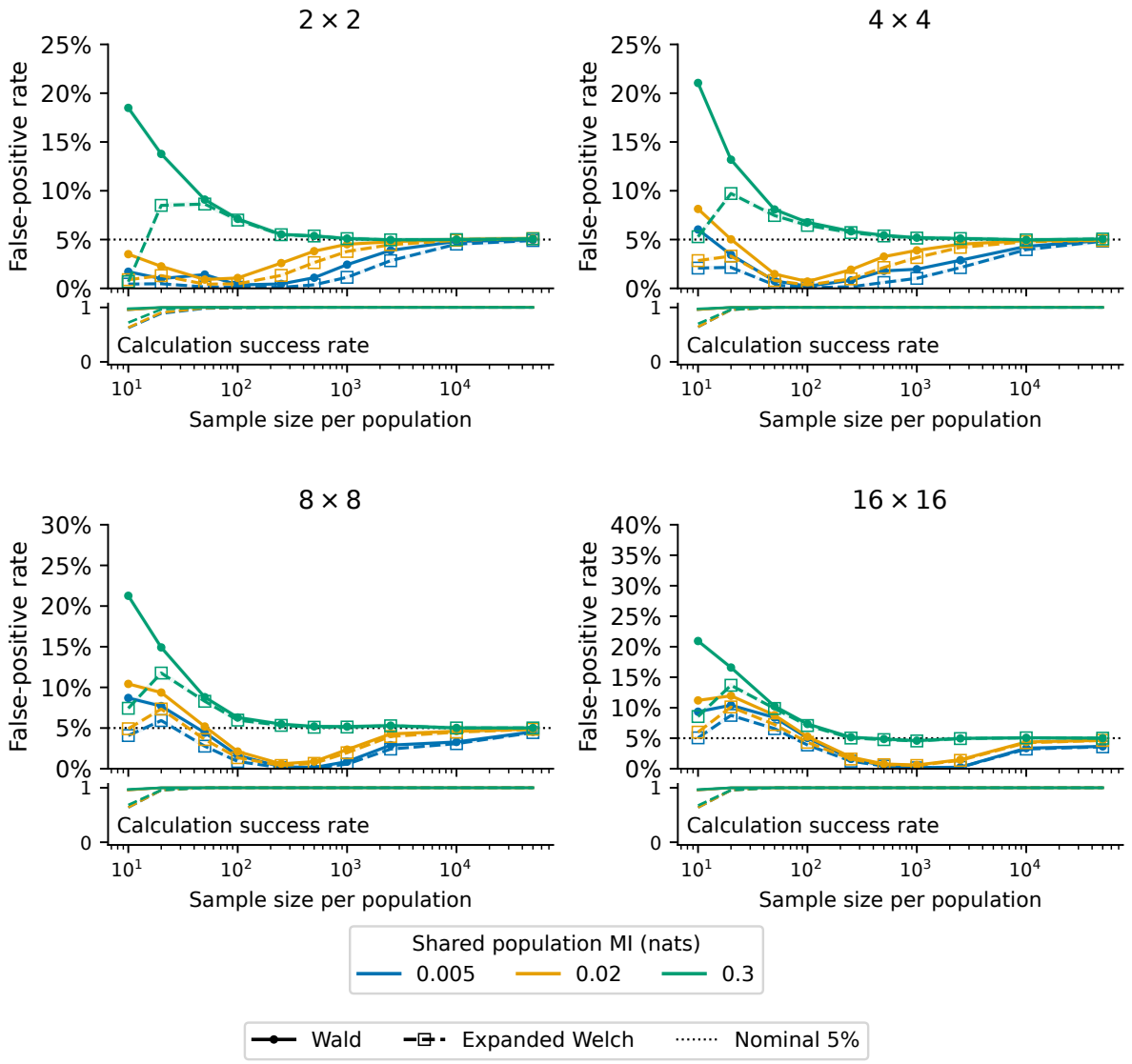


FIGURE 6.5: False-positive rates across shared MI values under same-skew margins.

The weaker shared MI values produce deep conservative ranges, with the trough occurring later as table size increases. Uneven margins leave more category combinations rarely observed, prolonging errors in the estimated cell probabilities and standard error. With identical population tables, the expected bias of the MI difference still cancels, so it cannot explain these departures. The diagnostic comparison below shows how an oversized standard error can shrink the statistic and suppress rejection. At shared MI 0.30, both methods instead over-reject at several small sample sizes before approaching 5% with more observations.

At the smallest sample sizes, the lower strips show that calculations for the Expanded Welch test can succeed less often than those for the Wald test, so failed calculations also lower the Expanded Welch test's rejection rate when all repetitions are counted. However, the conservative troughs persist even when both tests' calculations succeed in every repetition, so calculation failure is not their sole explanation. At larger samples, the methods' curves become close as the estimated degrees of freedom increase, but both false-positive rates can remain below 5% in the weak-association, larger-table settings.

6.2.3 Different-skew margins

Figure 6.6 uses $m_k(0.7)$ for P and $m_k(0.8)$ for Q. The population MI values are equal, but the tables differ, allowing their MI estimates to have different residual biases that can shift the shared statistic away from zero under the null.

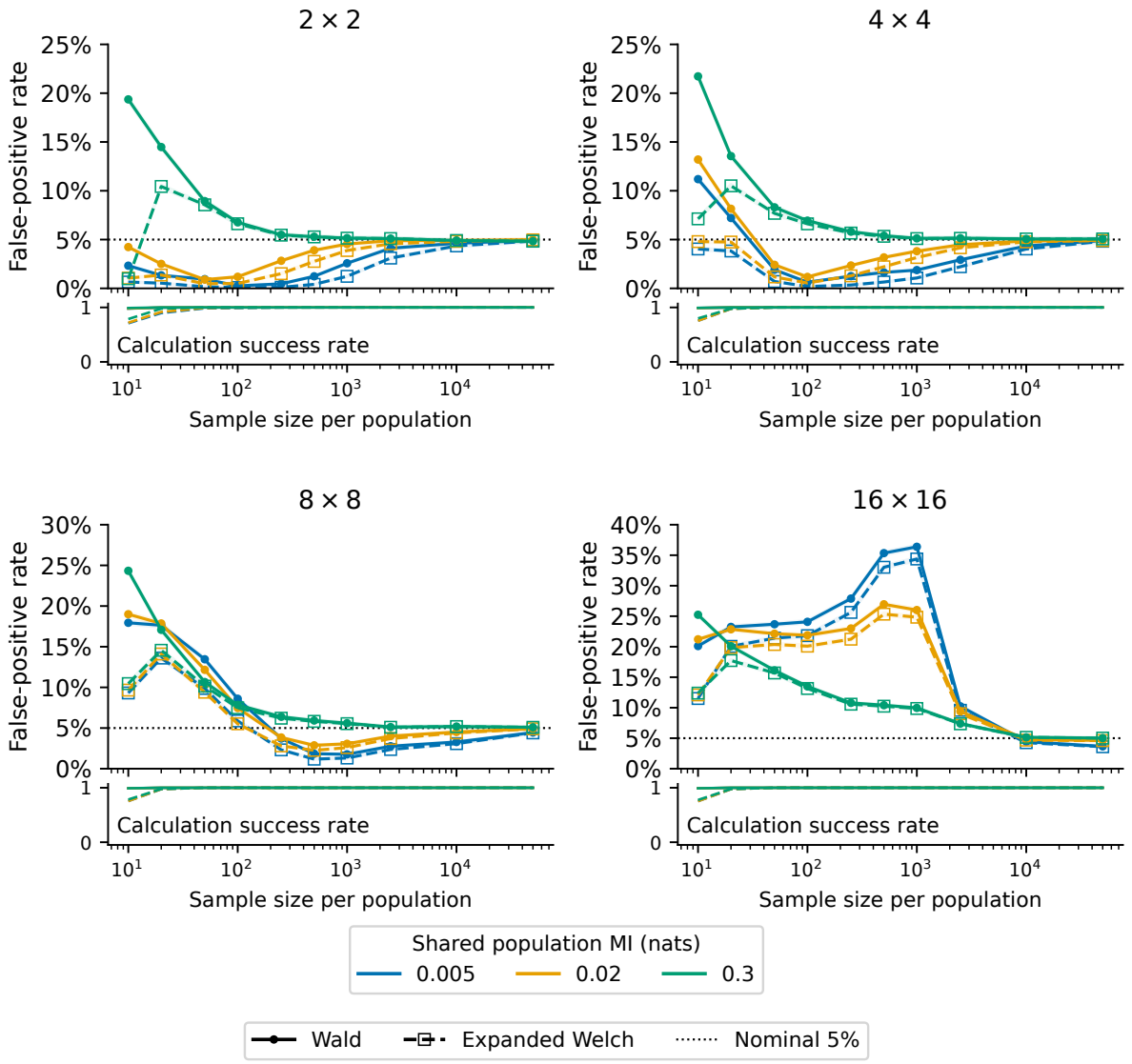


FIGURE 6.6: False-positive rates across shared MI values under different-skew margins.

The two smaller tables retain the conservative ranges at weaker MI and the excessive rejection at stronger MI seen in the other margin regimes. The 16×16 panel shows a more severe failure: at shared MI 0.005 and $n = 1{,}000$, the Wald test rejects in 36.38% of repetitions and the Expanded Welch test rejects in 34.38%. The larger cutoff reduces rejection but leaves it far above 5%. Both methods' calculations succeed in every repetition, so these high rates reflect inaccurate inference rather than calculation failures.

The contrast across shared MI depends on the bias relative to sampling spread. At 16×16 and $n = 1{,}000$, the bias in the estimated MI difference is about 0.03 nats at all three MI levels, but the sampling standard

deviation is larger at stronger association. The shift therefore has a greater effect relative to the spread at weaker MI, accompanying the much higher rejection there.

With sufficiently large samples, bias becomes small relative to the sampling spread and excessive rejection subsides. At shared MI 0.30, the large-sample curves settle near 5%, whereas the weakest-MI 16×16 case eventually falls below the target. An oversized standard error can still suppress rejection after the bias has diminished, taking the rate below the target.

6.2.4 Explaining the main departures from 5%

Table 6.1 contrasts three settings with a shared MI of 0.02 nats to illustrate different estimation errors. Because the population MI difference Δ is zero, "Bias / SD" equals the sampling mean of $\widehat{\Delta}_{\mathrm{BC}}$ divided by its observed sampling standard deviation. The variance ratio divides the average $\widehat{\mathrm{SE}}^2$ by the observed variance of $\widehat{\Delta}_{\mathrm{BC}}$ about its sampling mean, so it measures uncertainty estimation separately from bias. Ratios below one indicate underestimation and ratios above one indicate overestimation. Both methods' calculations succeed in every repetition in these three settings. Rates are percentages of all 100,000 repetitions.

TABLE 6.1: Estimation errors and false-positive rates at three illustrative null settings.

Setting	Bias / SD	Variance ratio	Wald (%)	Expanded Welch (%)
Uniform 4×4, $n = 20$	0.003	0.827	8.35	4.90
Same skew 16×16, $n = 1{,}000$	-0.003	1.929	0.60	0.53
Different skew 16×16, $n = 1{,}000$	2.058	1.938	26.01	24.84

In the uniform 4×4 setting, bias in the estimated MI difference is negligible, but the average squared standard error understates the observed sampling variance. For a given estimated difference, a smaller denominator enlarges $|T|$ and tends to increase rejection. The Expanded Welch test's larger cutoff reduces that excess and brings the rate close to 5%, showing how the cutoff can offset an error in the shared standard error at a particular population pair.

The two 16×16 settings have almost the same variance ratio but opposite rejection behaviour. Under same-skew margins, the biases in the two MI estimates cancel and the oversized standard error suppresses rejection, which the Expanded Welch test lowers further. Under different-skew margins, the positive bias

places the sampling mean of the estimated difference more than two sampling standard deviations above zero. Changing the cutoff leaves that bias in the estimated MI difference unchanged, helping explain why both methods reject far too often even though the estimated standard error is oversized.

Across the grid, the two tests' false-positive rates become close to one another with larger samples, but both can remain below the 5% target. For example, at shared MI 0.005 and $n = 50{,}000$, both tests have false-positive rates below 4% in every 16×16 margin regime, with the average squared standard error still exceeding the observed sampling variance of the estimated MI difference. The Expanded Welch test improves calibration where its lower rejection corrects an excess, while increasing conservatism where the Wald test already rejects too rarely.

6.2.5 Comparison with ordinary Welch degrees of freedom

To check whether MI-specific degrees of freedom make a difference, we also use a reference distribution with ordinary Welch degrees of freedom at three illustrative null settings. This comparison keeps the same estimated MI difference, standard error, and statistic, but substitutes component degrees of freedom $n_P - 1$ and $n_Q - 1$ into Equation (4.8). These sample-size-based values come from the comparison of sample means; for MI they serve as a benchmark rather than a derivation of the variance estimate's sampling variation. The Expanded Welch test instead uses the component degrees of freedom derived in Chapter 4.

The comparison reuses the same 100,000 sampled pairs per setting as the main null study, with shared MI 0.02 nats and equal sample sizes $n_P = n_Q = n$. All three methods' calculations succeed in every repetition.

TABLE 6.2: False-positive rates with ordinary and MI-specific Welch degrees of freedom.

Setting	Wald (%)	Ordinary Welch (%)	Expanded Welch (%)
Uniform 4×4, $n = 20$	8.35	7.35	4.90
Uniform 4×4, $n = 50$	4.87	4.57	3.24
Different skew 16×16, $n = 1{,}000$	26.01	25.95	24.84

At $n = 20$, using ordinary Welch degrees of freedom reduces excessive rejection less than the Expanded Welch test does. At $n = 50$, the Wald test's false-positive rate is already close to 5%, and the Expanded

Welch test instead produces the most conservative rate. In the different-skew 16×16 case, neither degrees-of-freedom choice removes the excessive rejection associated with bias in the shared MI difference. The MI-specific calculation therefore changes the results beyond merely replacing the normal distribution with a Student t distribution, but the larger reduction in rejection does not consistently improve calibration.

6.3 RQ3: Power across MI differences

The power study repeats the four table sizes, three margin regimes, and ten sample sizes of the broad null comparison at three MI-gap magnitudes: $|\Delta| = 0.005, 0.01, 0.02$ nats. Population P has MI 0.02 nats, while Q has MI 0.025, 0.03, or 0.04 nats, respectively, so each signed difference $\Delta = I(P) - I(Q)$ is negative. Each plotted rate divides the number of rejections by all 100,000 repetitions at that setting, with unsuccessful calculations counted as non-rejections. Shading gives pointwise 95% intervals, and the lower strips show the calculation success rates.

At the usual p-value threshold of 0.05, the Expanded Welch test's larger cutoff makes its rejections a subset of those from the Wald test, so the comparison quantifies how much detection probability is lost. These rates are read alongside the false-positive results in Section 6.2: high rejection under an alternative can coexist with excessive rejection under equal MI. Because the tests are two-sided, power also includes rejections whose estimated MI difference has the opposite sign to the population difference.

6.3.1 Uniform margins

Figure 6.7 uses uniform row and column margins in both populations. Each panel increases the shared sample size along the logarithmic horizontal axis, while colour identifies the MI-gap magnitude. Solid lines with filled circles show the Wald test's power, and dashed lines with open squares show that of the Expanded Welch test. The same colour and line conventions apply to all three power figures, with overlapping method curves indicating equal or visually indistinguishable power. Curves for different MI gaps also meet near 100% once almost every repetition rejects the null hypothesis.

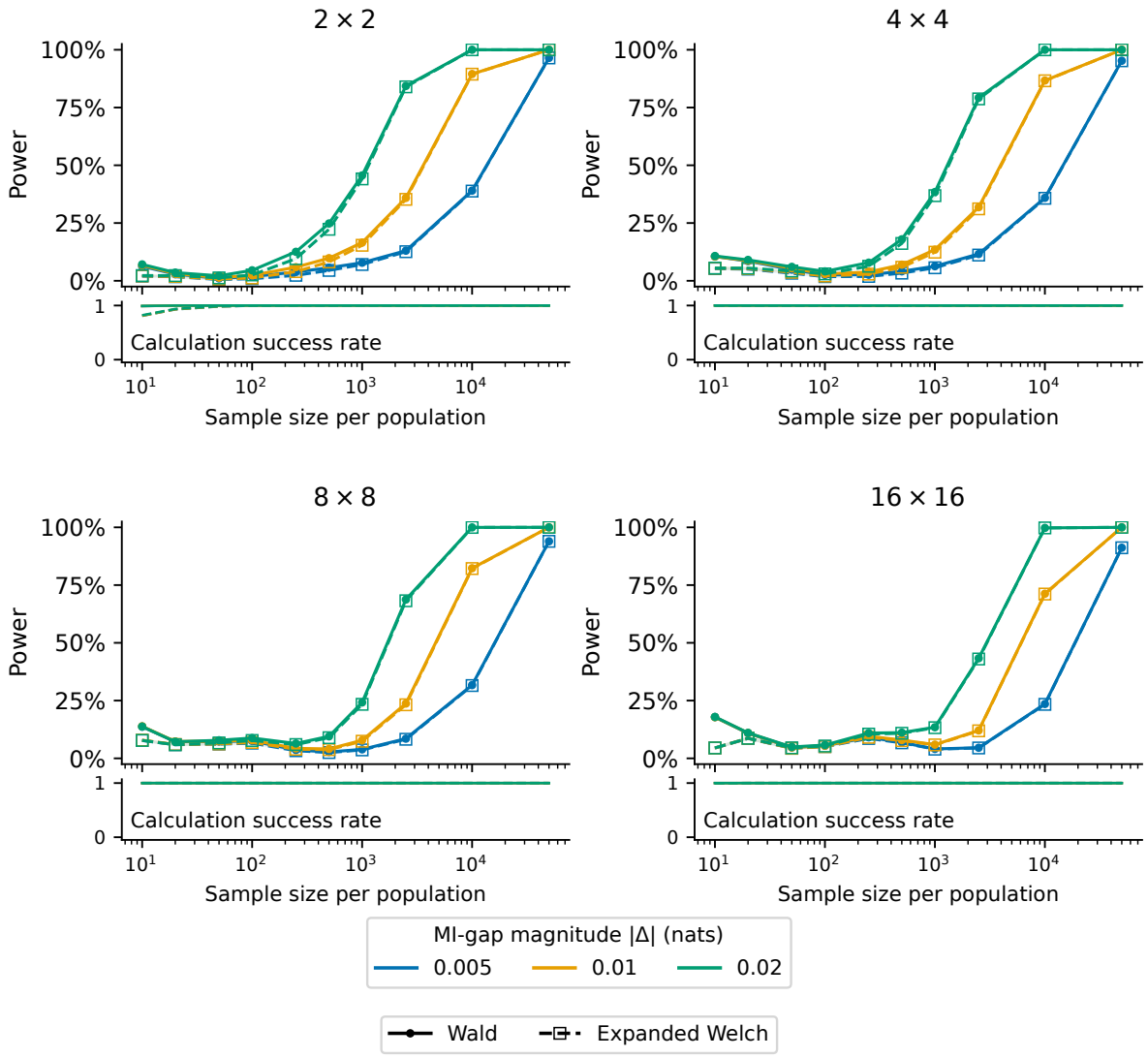


FIGURE 6.7: Power across three MI differences under uniform margins.

For each MI gap, all four panels show low detection over part of the small-sample range, followed by a sustained rise with more observations. The early shoulders and troughs also occur in the null rejection curves, where errors in the estimated standard error and a poor fit between the sampling distribution of T and its reference distribution can produce excessive or conservative rejection. Consequently, the early high points are not evidence that fewer observations reliably identify the MI difference.

Larger tables require more observations before power becomes high. For the 0.01-nat gap at $n = 10{,}000$, the Expanded Welch test detects the difference in 89.44% of 2×2 repetitions and 71.12% of 16×16 repetitions. This ordering reflects both the sparse-table estimation errors discussed in Section 6.2.4 and

the population variances: in this copying family, increasing table size also increases $V(P)$ and $V(Q)$. Their contributions $V(P)/n + V(Q)/n$ therefore give a larger first-order sampling variance for the same MI gap and sample size. As the variance decreases with more observations, the estimated difference becomes more consistently separated from zero and power rises.

Once samples are sufficiently large, increasing the MI gap shifts the power curves upwards. In the 4×4 panel at $n = 2{,}500$, the Expanded Welch test's power rises from 11.14% for the 0.005-nat gap to 78.74% for the 0.02-nat gap. Increasing $I(Q)$ also increases $V(Q)$, but in these populations the MI gap grows faster than the first-order sampling standard deviation of the difference. The true MI gap is therefore larger relative to the sampling spread of the estimated difference.

For every MI gap, power is no higher for the Expanded Welch test than for the Wald test. For the 0.01-nat gap, the largest power difference occurs at the smallest sample size in the 16×16 panel. The Expanded Welch test's calculation success rate at that setting is above 99%, so most of that separation comes from the larger cutoff rather than failed calculations. The difference in power between the tests becomes small in the larger-sample range as increasing degrees of freedom bring the Student t distribution close to the standard normal distribution.

6.3.2 Same-skew margins

Figure 6.8 assigns both populations the row and column profile $m_k(0.8)$, retaining the same three MI gaps and sample sizes. The margins agree between populations, but their different MI values require different copying probabilities and hence different population tables.

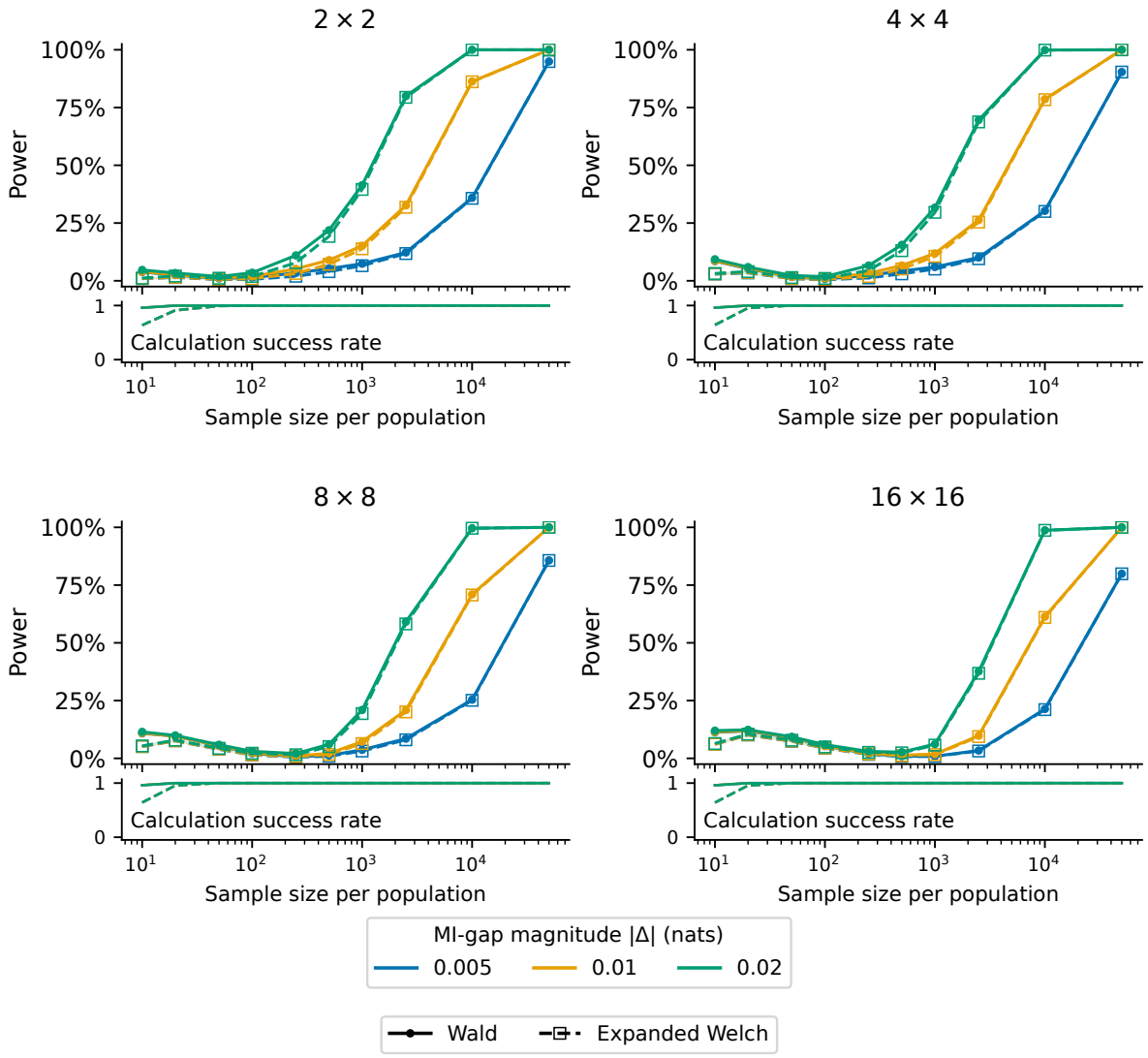


FIGURE 6.8: Power across three MI differences under same-skew margins.

The curves again have an early low-power range followed by recovery, with the trough occurring later for larger tables. Relative to uniform margins, the uneven frequencies leave more combinations rarely observed, and the population pointwise-MI variances are also larger. Both features make a given gap harder to detect. At the smallest sample sizes, the lower strips show that failed calculations for the Expanded Welch test also reduce its power when all repetitions are counted, so the larger cutoff is not the only reason for its lower power. For the 0.01-nat gap in the 16×16 case at $n = 1{,}000$, the average squared standard error is 1.81 times the observed variance of the difference, while bias in the estimated

MI difference is small relative to the true gap. The oversized denominator suppresses $|T|$, contributing to the Expanded Welch test's power of only 1.64% at that setting.

Both methods gain power once larger samples improve the cell estimates and reduce the spread of the MI difference. For the 0.01-nat gap, the Expanded Welch test's power reaches 60.99% in 16×16 at $n = 10{,}000$, compared with 71.12% under uniform margins. The smaller 0.005-nat gap remains harder to detect even at $n = 50{,}000$, where the Expanded Welch test's power reaches 79.87% in 16×16, compared with over 99.9% for both larger gaps. The differences between methods become much smaller than these differences between table sizes and margin regimes.

6.3.3 Different-skew margins

Figure 6.9 uses $m_k(0.7)$ for P and $m_k(0.8)$ for Q. Each true difference is negative, but the unequal margins can give the MI estimates different residual biases, shifting the centre of their difference.

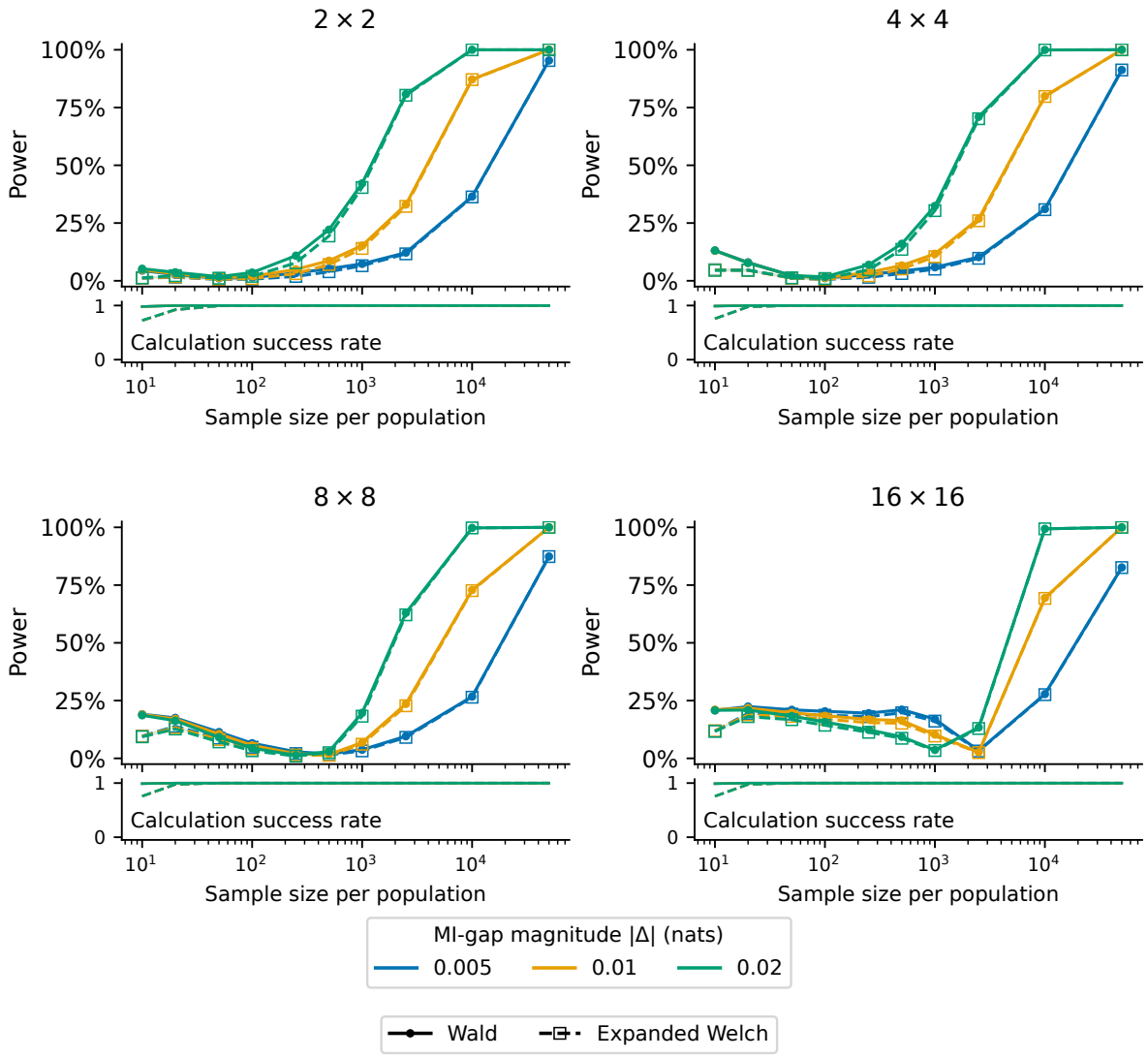


FIGURE 6.9: Power across three MI differences under different-skew margins.

The 2×2 and 4×4 panels retain the low-power range and subsequent rise, while the two larger tables have more pronounced early rejection before falling and recovering. The 16×16 curve shows why this early rejection needs careful interpretation. For the 0.01-nat gap at $n = 1{,}000$, the estimated MI difference has positive bias of approximately 0.0307 nats, moving its sampling mean from the true -0.01 to approximately $+0.0207$ nats. The Expanded Welch test rejects in 9.66% of repetitions, but more than 99.9% of those rejections have a positive estimated difference, opposite to the true direction. The shared biased numerator therefore produces rejection without reliably identifying which population has greater MI, as in the displaced null comparison in Section 6.2.4.

This positive bias also explains why a larger MI gap can give lower rejection at the same sample size. In 16×16 at $n = 1{,}000$, the Expanded Welch test's power falls from 16.11% for $|\Delta| = 0.005$ to 3.39% for $|\Delta| = 0.02$. The positive bias remains about 0.03 nats across these gaps, so making the true difference more negative moves the sampling centre closer to zero. Over this range, the decrease in rejections with positive estimated MI differences outweighs the increase in rejections with negative estimated MI differences. The ordering reverses at larger samples, once bias is small enough for most rejections to have an estimated MI difference with the same negative sign as the population difference.

Along the 0.01-nat curve, increasing sample size reduces the positive bias and moves the sampling centre back towards zero. At $n = 2{,}500$, the remaining positive bias nearly cancels the negative population difference, leaving the sampling mean close to zero, and the Expanded Welch test's power falls to 2.48%. With still more observations, the bias diminishes further and the sampling centre approaches the negative population difference, while sampling variance decreases. Rejections with negative estimated MI differences then become more frequent and power becomes high. Changing the cutoff lowers rejection on both sides of the trough without changing the bias in the estimated MI difference.

Across all three margin regimes, both methods exceed 99.9% power at $n = 50{,}000$ for gaps of 0.01 and 0.02 nats, whereas power for the 0.005-nat gap ranges from 79.87% to 96.47%. At $n = 10{,}000$, the method power differences are below 0.32 percentage points across all three gaps, table sizes, and margin regimes. Thus the gap magnitude strongly affects detection once the estimates become accurate, while the Expanded Welch test's larger cutoff gives a smaller power loss relative to the Wald test in this larger-sample range.

6.4 Example application to online shopping data

This example illustrates how the Wald test and the Expanded Welch test are applied and interpreted in practice by comparing month–purchase and weekend–purchase associations between new and returning visitors. The Online Shoppers Purchasing Intention dataset records website sessions and whether each session ends in a purchase (Sakar and Kastro, 2018). For each predictor, X is month or weekend status and Y is the purchase outcome, giving joint distributions P and Q of (X, Y) among new and returning visitors, respectively. Each comparison therefore tests $H_0 : I(P) = I(Q)$ for that predictor and purchase outcome.

The dataset contains 12,330 sessions, each belonging to a different user according to the source documentation (Sakar and Kastro, 2018). We retain all 1,694 sessions labelled `New_Visitor` and 10,551 labelled `Returning_Visitor`, excluding the 85 `Other` sessions. The outcome is the original Boolean `Revenue` variable, which is true when a purchase occurs and false otherwise. We treat the two visitor groups as independent samples and the sessions within each group as independent observations.

January and April have no recorded sessions, so each month table has 10×2 cells, with rows for the ten recorded months and columns for no purchase or purchase. All ten months occur in both visitor groups, although the new-visitor February row contains only one session and no purchase. Both groups use the same categories, retaining zero cells without adding artificial counts to empty cells. Replacing month with the Boolean `Weekend` variable gives a 2×2 table for each group, with weekday and weekend rows and all four cells observed.

The corrected difference $\widehat{\Delta}_{\mathrm{BC}}$ is new-visitor MI minus returning-visitor MI, so a negative value indicates higher estimated MI among returning visitors. The leading MI bias correction subtracts $(10-1)(2-1)/(2n) = 9/(2n)$ from each month plug-in estimate (Miller, 1955; Moddemeijer, 1989), and $1/(2n)$ from each weekend estimate, where n is the relevant group size. Table 6.3 gives the corrected MI estimates, their differences, and shared standard errors, $\widehat{\mathrm{SE}}$, all in nats, alongside the resulting $T = \widehat{\Delta}_{\mathrm{BC}}/\widehat{\mathrm{SE}}$.

TABLE 6.3: Corrected MI estimates and uncertainty of their differences.

	Corrected MI				
Predictor	**New**	**Returning**	$\widehat{\Delta}_{\mathrm{BC}}$	$\widehat{\mathrm{SE}}$	T
Month	0.004316	0.017880	-0.013563	0.003333	-4.069
Weekend	0.000664	0.000727	-0.000063	0.001124	-0.056

For month, returning visitors have the higher corrected MI estimate, giving a negative difference whose magnitude is about 4.07 times its estimated standard error:

$$T \approx \frac{-0.013563}{0.003333} \approx -4.069. \tag{6.1}$$

Both tests use this statistic, but the Wald test compares it with the standard normal distribution, whereas the Expanded Welch test uses a Student t distribution with estimated degrees of freedom. Table 6.4 gives the Expanded Welch test's degrees of freedom and the unadjusted two-sided p-values, which we compare with the usual p-value threshold of 0.05.

TABLE 6.4: Two-sample test results for online shopping sessions.

		Unadjusted p-value	
Predictor	**Expanded Welch degrees of freedom**	**Wald**	**Expanded Welch**
Month	26.79	4.71×10^{-5}	3.73×10^{-4}
Weekend	2.18	0.9554	0.9601

Both methods reject equal month–purchase MI because their p-values are below 0.05. The estimates suggest that month provides more information about the purchase outcome among returning visitors than among new visitors. The Expanded Welch test gives a larger p-value because the Student t distribution with 26.79 degrees of freedom assigns more probability to statistics at least this far from zero than the standard normal distribution.

For weekend, both corrected MI estimates are approximately 0.0007 nats, and their difference is only about 0.056 times its estimated standard error. The statistic therefore lies close to zero under either reference distribution, so both p-values are near one and neither test rejects equal weekend–purchase MI. This non-rejection does not establish equality. The Expanded Welch test's 2.18 degrees of freedom remain small despite the large samples because $\widehat{V}(P)$ and $\widehat{V}(Q)$ have substantial estimated fluctuations relative to their magnitudes. Equation (4.44) compares each squared estimate with its estimated sampling variance, so greater relative uncertainty gives fewer component degrees of freedom, as illustrated near independence in Figure 6.3.

6.5 Discussion

The Expanded Welch test improves on the Wald test where its larger cutoff corrects excessive false-positive rates. For 4×4 tables with uniform margins at shared MI 0.02 nats and $n = 20$, false-positive rates are 8.35% for the Wald test and 4.90% for the Expanded Welch test (Table 6.1). This reduction towards 5% supports the motivation of Pan and Wall (2002): accounting for variability in an estimated variance can reduce excessive Wald rejection. However, where the Wald test already rejects too rarely, the Expanded Welch test lowers rejection further. The accompanying loss of power must therefore be interpreted alongside whether the reduction in false positives improves calibration.

Choosing the degrees of freedom requires approximations to the sampling mean and variance of the pointwise-MI variance estimates. Using V in place of a larger sampling mean of $\widehat{V}$ can put the population first-order degrees of freedom below the moment target even when the sampling variance of $\widehat{V}$ is predicted accurately.

The standard error is calculated from the variance estimates in each observed pair of tables. For a given estimated MI difference, an oversized standard error reduces $|T|$. Table 6.1 shows that, under same-skew 16×16 margins, the average squared standard error is about 1.93 times the observed variance of the MI difference. With negligible difference bias in that setting, a larger cutoff compounds the suppression of rejection. This agrees with Bell and McCaffrey (2002), who find that Satterthwaite degrees of freedom can make tests particularly conservative when the variance estimator has positive bias.

Bias in the MI difference connects the null failures to the misleading power patterns. With identical population tables and sample sizes, the biases of the two MI estimates cancel in their difference. With different margins, unequal biases can shift the estimated difference away from zero under equal MI. When population MI values differ, those biases can also give the estimated difference the opposite sign to the population difference. The residual information-estimation bias discussed by Paninski (2003) thus affects both false-positive rates and the estimated direction of a difference. These findings explain why high rejection under an alternative need not identify the population with greater MI reliably, and why agreement between the Wald test and the Expanded Welch test on observed data does not establish accurate inference. Within the tested settings, the larger cutoff corrects some excessive rejection, but reliable comparisons also require an estimated MI difference with small bias relative to its sampling spread and a standard error that describes that spread.

CHAPTER 7

Conclusions

7.1 Contributions and main findings

This thesis develops and evaluates a test for comparing mutual information (MI) between two categorical variables using independent samples from two populations. The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains the estimated MI difference and standard error used by the Wald test but replaces the standard normal reference distribution with a Student t distribution. The mathematical contribution is a first-order approximation to how the variance estimates underlying the standard error change between samples. Their mean and sampling variability determine the degrees of freedom, which control the rejection cutoff.

Across the tested populations, the Expanded Welch test brings some excessive false-positive rates closer to the 5% target, but lowers rejection further where the Wald test already rejects too rarely. The larger cutoff can reduce power, the probability of detecting a real MI difference, while leaving errors in the estimated MI difference and standard error used by both tests unchanged. In the online-shopping application, both tests reject equal month–purchase MI, with the higher corrected MI estimate among returning visitors. Neither test provides evidence of a difference in weekend–purchase MI.

7.1.1 RQ1: Accuracy of variance approximations

In the tested populations, the population first-order calculation gives degrees of freedom below the target calculated from repeated samples over much of the sample-size range. The calculation uses the population pointwise-MI variance in place of the larger average of the variance estimates from repeated samples. Squaring this smaller value contributes to the downward error even where the sampling variance of these estimates is predicted fairly accurately. The moment approximations improve as samples grow, although

larger tables require more observations. Replacing population probabilities with sample proportions introduces further error into this variance prediction.

When the two variables are nearly independent, sampling noise creates apparent association in the observed tables, producing substantial variation that the first-order approximation misses. The sample-based calculation retains this apparent association, so the average estimated component degrees of freedom remain several units above zero in the 4×4 experiment with 1,000 observations per population. The population first-order prediction instead approaches zero as MI decreases at this fixed sample size. The calculation in Appendix A.2 includes the additional sampling variation and explains this separation.

7.1.2 RQ2: False-positive rates under equal MI

In the uniform 4×4 setting in Table 6.1, an underestimated standard error contributes to excessive rejection, as in the regression examples reviewed in Section 2.4 (Bell and McCaffrey, 2002; Kauermann and Carroll, 2001). The Expanded Welch test's larger cutoff brings the false-positive rate closer to 5%. In the same-skew 16×16 setting, however, the standard error is too large and bias in the estimated MI difference is small, so the Wald test already rejects too rarely and the larger cutoff reduces rejection further. The different-skew 16×16 setting instead shows how bias moves the estimated difference away from zero even when the population MI values are equal.

Under different-skew 16×16 margins at shared MI 0.005 nats and 1,000 observations per population, false-positive rates are 36.38% with the Wald test and 34.38% with the Expanded Welch test, far above the 5% target. Both methods' calculations succeed in every repetition, so these high rates reflect inaccurate inference rather than failed calculations. With 50,000 observations per population at the same MI level, both tests give similar rejection rates below 5% in all three 16×16 margin regimes.

7.1.3 RQ3: Power across MI differences

At the usual 5% significance level, the Expanded Welch test cannot reject where the Wald test would not, so it cannot have higher power under these same testing rules. At the smallest samples, some calculations for the Expanded Welch test also fail, contributing to its lower reported power. With 10,000 observations per population, the power difference between the two tests is below 0.32 percentage points across all tested settings.

In sparse tables, bias can make the estimated difference point to the wrong population as having greater MI, even when the test rejects equality. This bias can also oppose the population difference so that increasing the true MI gap initially moves the average estimate closer to zero and reduces rejection. With sufficiently large samples, the bias diminishes and larger MI gaps become easier to detect.

7.2 Limitations

The main derivation assumes independent categorical observations, a fixed set of categories, and positive probability for every category combination. The degrees-of-freedom calculation also requires positive pointwise-MI variances and positive variances of the observation-level sensitivities in both populations. Paired samples would require accounting for dependence between them, while continuous variables or a growing number of categories would require extending the present derivation. At exact independence between the two variables, the first-order sampling term vanishes, leaving the quadratic sampling variation examined in Appendix A.2.

The experiments use one population construction, which increases the probability of matching categories while keeping category frequencies fixed and equal for rows and columns within each table. Although the experiments vary selected table sizes and MI values, they do not examine other association patterns or unequal row and column frequencies. The power study further restricts the comparisons by fixing one population's MI and only increasing the other's, so other starting MI values and the reverse direction of change remain untested.

Because the simulations reported here use equally sized samples, the same leading bias correction is subtracted from each MI estimate and cancels from their difference, leaving the correction's performance with unequal sample sizes untested. The two MI estimates can nevertheless have different remaining biases, so their difference can still be biased even when sample sizes are equal.

Both tests are evaluated at the usual 5% significance level, with confidence intervals quantifying simulation uncertainty at individual population and sample-size settings. Because the tests' actual false-positive rates can differ, the power comparison describes detection under these usual testing rules rather than at the same observed false-positive rate. False-positive rates can be assessed in the simulations because the population MI values are known. In the online-shopping application, the population MI values are unknown, so the example illustrates interpretation on observed data without assessing the tests' false-positive rates.

The main empirical comparison uses the Wald test, with ordinary Welch degrees of freedom examined at three illustrative null settings. These comparisons do not establish how the Expanded Welch test performs against resampling procedures or tests using other MI estimators.

7.3 Future work

Future work should first examine whether improved estimates of the MI difference and standard error reduce the departures observed here. Changing one estimator at a time would help distinguish their effects on false-positive rates, power, and the estimated direction of the MI difference.

Retaining terms omitted by the first-order approximation could improve the predicted average of the variance estimates used in the degrees-of-freedom calculation. Figure 6.3 shows that the error in this predicted average changes with MI and category frequencies, so a correction should account for the population table and be evaluated through its effects on false-positive rates and power. Near independence, the reference distribution for the MI difference could also retain the variation from sampling noise that the first-order approximation misses. Any resulting test must allow populations with different probability tables to satisfy the null hypothesis when their MI values are equal.

Broader experiments could test whether the findings persist across other starting MI values, either population having greater MI, unequal sample sizes, and other patterns of category frequencies and association. Within these comparisons, simulations based on plausible application populations could also examine how rare category combinations affect the results and how the tests behave when distinct populations have equal MI.

The derived degrees of freedom bring some excessive false-positive rates closer to 5%, at the cost of reduced power. The remaining failures show the limits of the Expanded Welch test's cutoff adjustment and motivate improving the estimates of the MI difference and its standard error in sparse settings.

APPENDIX A

Supporting MI Derivations

The first two calculations support the near-independence results in Chapter 4 and their interpretation in Chapter 6. The first calculation varies population association while keeping the margins fixed. The second retains the quadratic sampling variation needed at independence and very small MI. The final section explains why positive MI and positive pointwise-MI variance are equivalent when every population cell has positive probability.

A.1 Fixed-margin behaviour near independence

Let $P_0 = (q_{ij})$ be a joint distribution in which X and Y are independent, with positive cells $q_{ij} = a_i b_j$. Consider $P_t = P_0 + tH$, where every row and column sum of H is zero. For sufficiently small t, all cells remain positive and the margins remain a and b. We expand around $t = 0$, where $I(P_0) = 0$.

For $f(t) = I(P_t)$, the second-order Taylor formula is

$$f(t) = f(0) + f'(0)t + \frac{f''(0)}{2!}t^2 + O(t^3). \tag{A.1}$$

Here $O(t^3)$ collects the omitted terms whose magnitude is bounded by a constant times $|t|^3$ as t approaches zero. These terms become smaller than the retained quadratic term. The same notation below describes the size of each omitted remainder near zero.

The fixed margins make the pointwise MI in cell (i, j)

$$\ell_{P_t}(i,j) = \log\left(\frac{q_{ij} + tH_{ij}}{q_{ij}}\right) = \log\left(1 + \frac{tH_{ij}}{q_{ij}}\right) = \frac{tH_{ij}}{q_{ij}} - \frac{t^2 H_{ij}^2}{2q_{ij}^2} + O(t^3), \tag{A.2}$$

using $\log(1+z) = z - z^2/2 + O(z^3)$ with $z = tH_{ij}/q_{ij}$. Multiplying by the cell probability gives

$$\begin{aligned}(q_{ij} + tH_{ij})\ell_{P_t}(i,j) &= (q_{ij} + tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3)\right\} \\ &= tH_{ij} - \frac{t^2H_{ij}^2}{2q_{ij}} + \frac{t^2H_{ij}^2}{q_{ij}} + O(t^3) \\ &= tH_{ij} + \frac{t^2H_{ij}^2}{2q_{ij}} + O(t^3).\end{aligned}$$

Multiplication by q_{ij} supplies the negative quadratic term; multiplication by tH_{ij} supplies the positive one. Their combination leaves half the positive term. When we sum over cells, the first-order terms cancel because $\sum_{i,j} H_{ij} = 0$. Thus $f(0) = f'(0) = 0$, $f''(0) = \sum_{i,j} H_{ij}^2/q_{ij}$, and

$$I(P_t) = \frac{t^2}{2}\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.3}$$

To calculate $V(P_t)$, first square the pointwise-MI expansion:

$$\ell_{P_t}(i,j)^2 = \left\{\frac{tH_{ij}}{q_{ij}} + O(t^2)\right\}^2 = \frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3).$$

Multiplying by the cell probabilities and summing gives its second moment:

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y)^2\} &= \sum_{i,j}(q_{ij} + tH_{ij})\left\{\frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3)\right\} \\ &= t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned}$$

MI itself begins with a quadratic term, so $I(P_t)^2 = O(t^4)$. Subtracting it from this second moment leaves the quadratic term unchanged:

$$V(P_t) = t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.4}$$

The row conditional mean also has no first-order term. Its denominator remains a_i because the margins are fixed, and

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y) \mid X = i\} &= \frac{1}{a_i}\sum_{j}(q_{ij} + tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}} + O(t^2)\right\} \\ &= \frac{t}{a_i}\sum_{j}H_{ij} + O(t^2) = O(t^2).\end{aligned}$$

The last equality uses the zero row sum of H. The column mean has the same cancellation, using $\sum_i H_{ij} = 0$.

In Equation (4.30), the squared centred pointwise MI, $V(P_t)$, the conditional means, and $I(P_t)$ all begin at order t^2. Only the unsquared pointwise-MI term in the bracket has a first-order contribution. Therefore Equation (4.30) supplies the first-order term:

$$g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.5}$$

Since g_{P_t} has mean zero, its variance is its second moment:

$$\begin{aligned}\tau^2(P_t) &= \sum_{i,j}(q_{ij} + tH_{ij})\left\{\frac{2tH_{ij}}{q_{ij}} + O(t^2)\right\}^2 \\ &= 4t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned} \tag{A.6}$$

Comparing the leading terms of $I(P_t)$, $V(P_t)$, and $\tau^2(P_t)$ gives $V(P_t) \approx 2I(P_t)$ and $\tau^2(P_t) \approx 4V(P_t)$ for small positive MI along this path. At $t = 0$, the first-order sampling variance is zero; Appendix A.2 treats that case separately.

A.2 The independence boundary

At independence, $p_{ij} = p_{i+}p_{+j}$, so every population pointwise-MI value is $\ell_P(i,j) = \log 1 = 0$. Consequently, $I(P) = V(P) = 0$, and the first-order change in MI is zero. Sampling can still produce positive estimated MI, but its leading variation comes from second-order cell errors.

To calculate this variation, fix a strictly positive population table P in which X and Y are independent, and let $h = (h_{ij})$ describe a redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define

$$P_\varepsilon = P + \varepsilon h, \qquad f(\varepsilon) = I(P_\varepsilon), \tag{A.7}$$

for ε small enough that every cell remains positive. The expansion point is $\varepsilon = 0$, where $f(0) = I(P) = 0$. The second-order Taylor formula gives

$$f(\varepsilon) = f(0) + f'(0)\varepsilon + \frac{f''(0)}{2!}\varepsilon^2 + o(\varepsilon^2). \tag{A.8}$$

The first derivative is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = -\sum_{i,j} h_{ij} = 0. \tag{A.9}$$

Here the pointwise-MI terms vanish at independence, and the remaining terms cancel because the probabilities must still sum to one.

The row and column changes are $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Applying $\mathrm{d}^2(u \log u)/\mathrm{d}u^2 = 1/u$ to the joint and marginal terms in Equation (2.11) gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \tag{A.10}$$

Thus the leading change is

$$I(P_\varepsilon) = \frac{\varepsilon^2}{2} \left\{ \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}} \right\} + o(\varepsilon^2). \tag{A.11}$$

The first sum measures squared cell changes, whereas the last two remove the contributions due to changed margins, leaving the change in association. The remainder is negligible relative to ε^2 as ε approaches zero.

For a sample from a multinomial distribution, take $\varepsilon h = \widehat{P} - P$. Since cell-probability errors typically have size proportional to $n^{-1/2}$, squaring them makes $\widehat{I}(P)$ proportional to n^{-1} under independence. The quadratic form above measures the part of these errors that changes association after accounting for the margins. An unrestricted joint table has $rc - 1$ free probabilities because its cells sum to one. An independent table is determined by $r - 1$ row probabilities and $c - 1$ column probabilities. The number of remaining association directions is therefore

$$d = (rc - 1) - \{(r - 1) + (c - 1)\} = (r - 1)(c - 1). \tag{A.12}$$

For example, start with a 2×2 independence table whose four cell probabilities are all $1/4$. Keeping every row and column probability equal to $1/2$ allows tables of the form

$$\begin{pmatrix} 1/4 + \varepsilon & 1/4 - \varepsilon \\ 1/4 - \varepsilon & 1/4 + \varepsilon \end{pmatrix}, \qquad |\varepsilon| < 1/4.$$

The diagonal cells gain ε, while the off-diagonal cells lose ε, so every row and column total is unchanged. Conversely, once the change in the first cell is chosen, preserving its row total forces the opposite change

in the second cell. Preserving the first column total and the second row total then determines the other two changes. There is therefore only one free amount of association change, illustrating $d = 1$.

For this uniform independence table, one standardised sampling coordinate is

$$\sqrt{n}\,(\widehat{p}_{11} - \widehat{p}_{12} - \widehat{p}_{21} + \widehat{p}_{22})\,.$$

The corresponding contrast of population probabilities equals 4ε for the table above, so it records the diagonal-versus-off-diagonal change. In a sample, this contrast is the average of observation contributions that equal $+1$ when the categories match and -1 when they differ. Under uniform independence, these two outcomes each have probability $1/2$, giving mean zero and variance one. The central limit theorem therefore gives the scaled contrast a standard normal limiting distribution.

For general $r \times c$ tables with fixed dimensions and positive cell probabilities, the multinomial central limit theorem approximates the scaled cell errors by a multivariate normal distribution. Linear combinations of these standardised errors separate changes in the margins from changes in association. After removing the margin changes, the remaining d association coordinates can be chosen to have independent standard normal limiting distributions. The quadratic form for $2n\widehat{I}(P)$ is then their sum of squares, giving the likelihood-ratio result

$$2n\widehat{I}(P) \xrightarrow{d} \chi^2_{(r-1)(c-1)} \tag{A.13}$$

(Wilks, 1938; Brillinger, 2004). This is the usual one-table likelihood-ratio, or G, test of independence. Its chi-squared distribution comes from quadratic cell errors, whereas the Wald test and the Expanded Welch test use the first-order MI error. The two-sample first-order approximation therefore does not apply unchanged at exact independence.

Sampling variation at small positive MI.

The quadratic approximation also explains why estimated degrees of freedom remain several units when population MI is almost zero. Keep r and c fixed and let a positive population table approach independence with cell departures proportional to $n^{-1/2}$. The sampling errors, approximated by a multivariate normal distribution, and the population departure then contribute on the same scale. For this model based on a multinomial distribution, also assume strictly positive limiting cell probabilities. The association coordinates now include both sampling noise and the scaled population departure. Write $Z_1, \ldots, Z_d$ for independent variables with standard normal distributions and $\eta_1, \ldots, \eta_d$ for the population

shifts in those coordinates. The quadratic statistic is approximated by

$$\sum_{\ell=1}^{d}(Z_\ell + \eta_\ell)^2, \qquad \kappa = \sum_{\ell=1}^{d}\eta_\ell^2. \tag{A.14}$$

This sum has a noncentral chi-squared distribution, whose noncentrality κ is the squared size of the population shift. The limiting likelihood-ratio result under these local alternatives is given by Banerjee (2005).

To relate the population shift to MI, consider the approaching population P. Its difference from independence with the same margins is $p_{ij} - p_{i+}p_{+j}$, with zero row and column sums. Substituting this difference into Equation (A.11) makes the two marginal terms vanish and gives

$$2nI(P) \approx n\sum_{i,j}\frac{(p_{ij} - p_{i+}p_{+j})^2}{p_{i+}p_{+j}} \approx \kappa. \tag{A.15}$$

The weighted sum is the squared size of the scaled population association departure, explaining the approximation $\kappa \simeq 2nI(P)$. The MI in this local argument changes with n as the population approaches independence. Combining the noise and population shift gives the local approximation

$$G = 2n\widehat{I}(P) \mathrel{\dot\sim} \chi^2_d(\kappa), \qquad d = (r-1)(c-1), \qquad \kappa \simeq 2nI(P), \tag{A.16}$$

where $\mathrel{\dot\sim}$ denotes an approximate distribution. At independence, $\kappa = 0$, recovering Equation (A.13).

For a sampled table close to independence, apply the expansions in Appendix A.1 around the product of its empirical row and column probabilities. This gives

$$\widehat{V}(P) \approx 2\widehat{I}(P) = \frac{G}{n}, \qquad \widehat{\tau}^2(P) \approx 4\widehat{V}(P) \approx \frac{4G}{n}. \tag{A.17}$$

Substitution into the implemented component formula gives

$$\widehat{\nu}_P \approx \frac{2n(G/n)^2}{4G/n} = \frac{G}{2}. \tag{A.18}$$

Thus sampling noise contributes to the estimated degrees of freedom even when the population first-order approximation $nI(P)$ is very small.

The same approximation separates that plug-in calculation from matching the actual sampling moments. Under the approximation using a noncentral chi-squared distribution, $\mathrm{E}(G) \approx d + \kappa$ and $\mathrm{Var}(G) \approx 2(d + 2\kappa)$, giving

$$\mathrm{E}\{\widehat{V}(P)\} \approx \frac{d+\kappa}{n}, \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{2(d+2\kappa)}{n^2}. \tag{A.19}$$

Substituting these approximate sampling moments into the moment-matching formula gives

$$\nu_{P,\mathrm{mom}} \approx \frac{2\{(d+\kappa)/n\}^2}{2(d+2\kappa)/n^2} = \frac{(d+\kappa)^2}{d+2\kappa}. \tag{A.20}$$

When κ is much larger than d, this reduces to $\kappa/2 \simeq nI(P)$, as in Equation (4.52). When κ is small, however, the matched degrees of freedom approach d, while the mean plug-in degrees of freedom approach $d/2$.

For the 4×4, $n = 1{,}000$, $I(P) = 0.0001$ setting, $d = 9$ and $\kappa \simeq 0.2$. The local calculation gives mean estimated component degrees of freedom of approximately 4.6, rather than the population first-order value 0.1. It also predicts that the ratio of the average sample-based variance prediction to $\mathrm{Var}\{\widehat{V}(P)\}$ is approximately

$$\frac{4(d+\kappa)/n^2}{2(d+2\kappa)/n^2} = \frac{2(d+\kappa)}{d+2\kappa} \simeq 1.96. \tag{A.21}$$

This explains the scale of the estimated degrees of freedom in Figure 6.3 and the roughly twofold sample-based variance overprediction discussed alongside it. The population formula alone omits the quadratic sampling contribution responsible for both effects.

A.3 Positive MI and pointwise-MI variance

Assumption 4 in Chapter 3 states that positive MI is equivalent to positive pointwise-MI variance when every population cell has positive probability. For such a table $P = (p_{ij})$, the result is

$$I(P) > 0 \quad \Longleftrightarrow \quad V(P) > 0. \tag{A.22}$$

We establish this result by showing that zero variance forces zero MI, then considering the reverse direction. A probability-table example illustrates the two quantities, followed by a table showing why the positive-cell condition matters.

MI is an average; pointwise-MI variance measures spread.

Each cell has a pointwise-MI value that compares its probability with the probability under independence. MI averages these values, whereas $V(P)$ averages their squared departures from that average:

$$I(P) = \sum_{i,j} p_{ij} \ell_P(i,j), \tag{A.23}$$

$$V(P) = \sum_{i,j} p_{ij} \{\ell_P(i,j) - I(P)\}^2. \tag{A.24}$$

Zero variance means that the values do not differ across possible outcomes. In general, equal values can have a nonzero average, so constancy alone does not show that MI is zero. The additional restriction here comes from the cell probabilities having to sum to one.

From zero variance to zero MI.

Suppose $V(P) = 0$. Every term in Equation (A.24) is nonnegative. Because $p_{ij} > 0$ in every cell, a nonzero squared departure in even one cell would make the sum positive. The zero sum therefore requires

$$\ell_P(i,j) = I(P) \qquad \text{for every } (i,j). \tag{A.25}$$

Thus every cell has the same pointwise-MI value. To determine that value, substitute the definition of pointwise MI and exponentiate:

$$\log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right) = I(P), \tag{A.26}$$

$$\frac{p_{ij}}{p_{i+}p_{+j}} = e^{I(P)} = K, \tag{A.27}$$

$$p_{ij} = K p_{i+} p_{+j}. \tag{A.28}$$

The number K is the same in every cell because $I(P)$ is a single number for the whole table. It says that every actual cell probability is the same multiple of its probability under independence.

The products $p_{i+}p_{+j}$ form an independence table with the same margins as P. They already have total probability one:

$$\sum_i \sum_j p_{i+}p_{+j} = \sum_i p_{i+} \left(\sum_j p_{+j}\right) \tag{A.29}$$

$$= \sum_i p_{i+}(1) = 1. \tag{A.30}$$

The first step takes p_{i+} outside the inner sum because it does not depend on the column j. The column probabilities sum to one, and the remaining row probabilities also sum to one. Multiplying every entry of this reference table by K therefore multiplies its total probability by K. Since the actual table must also sum to one, Equation (A.28) gives

$$1 = \sum_{i,j} p_{ij} = K \sum_{i,j} p_{i+} p_{+j} = K. \tag{A.31}$$

Consequently $K = 1$, every cell satisfies $p_{ij} = p_{i+}p_{+j}$, and every pointwise-MI value is $\log 1 = 0$. Their weighted average is therefore $I(P) = 0$.

From zero MI to zero variance.

For the reverse direction, use the characterisation of zero MI in Chapter 2: $I(P) = 0$ implies independence, so $p_{ij} = p_{i+}p_{+j}$ in every cell. Each pointwise-MI value is then zero, giving

$$V(P) = \sum_{i,j} p_{ij} \{0 - 0\}^2 = 0. \tag{A.32}$$

The two directions establish $I(P) = 0$ if and only if $V(P) = 0$. MI is nonnegative and variance is nonnegative, so their positive cases are also equivalent, as stated in Equation (A.22).

A worked 2×2 example.

Consider the population table P and the independence table formed from its margins:

$$P = \begin{pmatrix} 0.4 & 0.1 \\ 0.1 & 0.4 \end{pmatrix}, \qquad P_{\text{ind}} = \begin{pmatrix} 0.25 & 0.25 \\ 0.25 & 0.25 \end{pmatrix}. \tag{A.33}$$

Every row and column totals 0.5, so each independence probability is $0.5(0.5) = 0.25$. A diagonal cell has actual-to-independence ratio $0.4/0.25 = 1.6$, giving pointwise MI $\log 1.6 \simeq 0.47000$. An off-diagonal cell has ratio $0.1/0.25 = 0.4$, giving pointwise MI $\log 0.4 \simeq -0.91629$. The two diagonal cells together have probability 0.8, and the other two have probability 0.2. Hence

$$I(P) = 0.8 \log 1.6 + 0.2 \log 0.4 \simeq 0.19274, \tag{A.34}$$

$$V(P) = 0.8\{\log 1.6 - I(P)\}^2 + 0.2\{\log 0.4 - I(P)\}^2 \simeq 0.30749. \tag{A.35}$$

MI is positive, and the differing pointwise-MI values give positive variance. The diagonal probabilities are larger than independence predicts, while the off-diagonal probabilities are smaller, preserving total

probability one. For P_{ind} itself, every ratio is one, so every pointwise-MI value is zero and both MI and the pointwise-MI variance are zero.

Why positive cell probabilities are required.

Now consider a population in which only the matching pairs can occur:

$$P = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}. \tag{A.36}$$

The margins are still 0.5 and 0.5. Each possible outcome has pointwise MI $\log(0.5/0.25) = \log 2$, so

$$I(P) = \log 2 > 0, \qquad V(P) = 0.5\{\log 2 - \log 2\}^2 + 0.5\{\log 2 - \log 2\}^2 = 0. \tag{A.37}$$

The zero-probability cells contribute no term to the variance, so the zero-variance condition constrains only the two possible outcomes. The ratio of two holds only for the diagonal cells, whose independence probabilities sum to 0.5. Doubling those probabilities gives total probability one without requiring a ratio of one in every cell. This table therefore shows that positive MI can coexist with zero pointwise-MI variance when some population cells are impossible. The positive-cell assumption concerns population probabilities, so an unobserved cell in a finite sample does not by itself violate the assumption.

APPENDIX B

Experimental Details

This appendix specifies the population construction, target-MI search, settings underlying the main results, sampling procedure, and outcome calculations. Together with the test formulas in Chapters 3–5, these details define the statistical experiment.

B.1 Copying construction and target selection

For the copying construction in Equation (5.1), summing a row gives

$$\sum_j p_{ij}(\lambda) = (1-\lambda)a_i \sum_j a_j + \lambda a_i = a_i. \tag{B.1}$$

Summing a column similarly gives a_j, so both margins are preserved. All cells are positive for $0 \leq \lambda < 1$.

MI increases continuously from zero to the entropy of a. With the margins fixed, differentiating the joint terms gives

$$\frac{\mathrm{d}^2 I\{P(\lambda)\}}{\mathrm{d}\lambda^2} = \sum_{i,j} \frac{(a_i \mathbf{1}_{\{i=j\}} - a_i a_j)^2}{p_{ij}(\lambda)} > 0. \tag{B.2}$$

The first derivative at zero is zero, because the change in probabilities sums to zero and all pointwise MI values are zero there. The positive second derivative makes the first derivative positive for $\lambda > 0$. At the endpoint, $Y = X$, so

$$I\{P(1)\} = -\sum_i a_i \log a_i. \tag{B.3}$$

These facts give a unique copying probability for each positive target below the entropy endpoint.

For each population, calculate this endpoint and solve $I\{P(\lambda)\} - I_{\text{target}} = 0$ on $[0, 1]$ using Brent's method (Brent, 1973). Absolute and relative root tolerances are 10^{-15} and 10^{-14}. The endpoint brackets the search; sampled populations use the interior solution. Accept a table only when every cell is positive,

its total and marginal errors are at most 10^{-12}, and its MI error is at most $10^{-12} + 10^{-8} I_{\text{target}}$. Every target is checked against its own population entropy endpoint.

B.2 Parameter grids and sampling

Table 5.2 specifies the parameter grids, with the null and power MI values given in Equations (5.6) and (5.7). Solve separately for each population's copying probability, since different margins can require different copying probabilities even under equal MI.

For one repetition, draw the counts from a multinomial distribution:

$$(N_{P,11}, \ldots, N_{P,kk}) \sim \text{Multinomial}(n_P; p_{11}, \ldots, p_{kk}), \tag{B.4}$$

and draw the corresponding Q counts independently. Apply both tests to each sampled pair of $k \times k$ tables, using 100,000 independent repetitions per distinct setting. The simulations use the PCG64 pseudo-random generator with master seed 20261021 for the approximation checks and null settings with a shared MI of 0.02 nats, and 20261101 for the other null settings and the power study. Separate streams distinguish settings and the two populations, while repeated settings share the same repetitions across displays.

All samples retain their declared table dimensions, including empty rows and columns. Use natural logarithms and $0 \log 0 = 0$, without adding artificial counts to empty cells. The bias correction for each plug-in MI is $(k-1)^2/(2n)$; do not truncate negative corrected estimates. Calculate the standard error using Equation (3.9) and the component and combined degrees of freedom using Chapter 4.

B.3 Calculation requirements and rejection rates

A calculation for the Wald test succeeds when it has a finite corrected difference, a finite positive standard error, $\widehat{V}(P) + \widehat{V}(Q) > 10^{-14}$, and a finite p-value. The threshold treats variances extremely close to zero as numerically unusable. The Expanded Welch test additionally requires positive finite degrees of freedom for both components and their combination, and a finite p-value for that test. If either component's degrees of freedom are not positive and finite, the calculation for the Expanded Welch test is unsuccessful.

The componentwise algorithm requires positive, finite degrees of freedom for both populations. If a sampled table has $\widehat{V} = \widehat{\tau}^2 = 0$, the component formula in Equation (4.44) requires $0/0$. The calculation

for the Expanded Welch test is unsuccessful even if the other table makes the shared standard error positive and finite.

The primary rejection rate divides rejections from successful calculations by all repetitions, with unsuccessful calculations counted as non-rejections. The study applies no minimum expected-count filter.

B.4 Bias, variance, and degrees of freedom

Let $\widehat{\Delta}_s$ be the corrected difference in repetition s. Its bias estimate is the mean of the finite-valued MI difference estimates minus the known Δ. Its observed sampling variance is

$$\frac{1}{M-1}\sum_{s=1}^{M}\left(\widehat{\Delta}_s - \frac{1}{M}\sum_{u=1}^{M}\widehat{\Delta}_u\right)^2, \tag{B.5}$$

where M counts the finite-valued differences, including those whose test calculation is unsuccessful. Centring at their sample mean separates variance from bias.

For the estimated / observed variance ratio, restrict the calculations to the same repetitions with a finite difference and finite squared standard error. Retain finite zero variance estimates. Divide the mean squared standard error by the observed variance on this matched subset. The ratio is defined only when the observed variance is positive.

Degrees-of-freedom summaries retain positive finite estimates. The mean-to-target ratios in Figure 6.2 are shown only when every component estimate is positive and finite. Finite zero variance estimates remain in moment calculations even when component degrees of freedom are undefined. These diagnostics do not isolate each estimation error's causal contribution to rejection.

Sampling variability of the estimated pointwise-MI variance.

For each selected population, measure the observed variability of $\widehat{V}(P)$ by replacing $\widehat{\Delta}_s$ with $\widehat{V}_s(P)$ in Equation (B.5). Compare the observed value with

$$\begin{aligned}\text{population prediction} &= \frac{\tau^2(P)}{n_P},\\ \text{mean plug-in prediction} &= \frac{1}{M}\sum_{s=1}^{M}\frac{\widehat{\tau}_s^2(P)}{n_P}.\end{aligned} \tag{B.6}$$

The population prediction uses the known probabilities, whereas the mean plug-in prediction uses the empirical probabilities in each repetition. Repeat both for Q. For each population, use the same repetitions with finite $\widehat{V}_s$ and $\widehat{\tau}_s^2$, including finite zeros and outputs whose test calculation is unsuccessful. Divide both predictions by the same observed variability, which must be positive.

The checks of cell contributions and the linear approximation accompanying Figure 6.1 use 10,000 of the approximation-study repetitions for each 16×16 different-skew population at shared MI 0.02 nats and sample sizes $n \in \{10, 20, 50, 100, 250, 1000, 10000\}$. Contributions to $\widehat{V}$ and $\widehat{\tau}^2$ are separated into once-observed cells whose row and column totals also equal one, other once-observed cells, and cells observed more than once.

For the linear-approximation check, calculate the first-order change in each repetition using the population sensitivity from Equation (4.30):

$$L_s(P) = \sum_{i,j} \{\widehat{p}_{s,ij} - p_{ij}\} g_P(i,j), \qquad R_s(P) = \widehat{V}_s(P) - V(P) - L_s(P). \tag{B.7}$$

Here $\widehat{p}_{s,ij}$ is the empirical probability in repetition s, $L_s(P)$ is the linear contribution, and $R_s(P)$ contains the remaining change. Since $\sum_{i,j} p_{ij} g_P(i,j) = 0$, the linear contribution is also the sample average of g_P in Equation (4.32). Its sampling variance is therefore $\tau^2(P)/n_P$. The decomposition $\widehat{V}_s(P) - V(P) = L_s(P) + R_s(P)$ is exact by definition of the remainder, even when the linear approximation is poor.

Taking variances and expanding the variance of this sum gives

$$\begin{aligned} \operatorname{Var}\{\widehat{V}_s(P)\} = {} & \operatorname{Var}\{L_s(P)\} + \operatorname{Var}\{R_s(P)\} \\ & + 2\operatorname{Cov}\{L_s(P), R_s(P)\}. \end{aligned} \tag{B.8}$$

Subtracting $V(P)$ does not change the variance because the population table is fixed. The linear contribution and remainder come from the same sample, so their covariance must be included. In particular, the difference between the full variance and the linear prediction is $\operatorname{Var}(R_s) + 2\operatorname{Cov}(L_s, R_s)$, which can be smaller than $\operatorname{Var}(R_s)$ when the covariance is negative.

Estimate the three terms from the same $M = 10{,}000$ repetitions using sample variances and covariance with denominator $M - 1$. Their sum equals the observed sample variance of $\widehat{V}_s$. For example, at $n = 50$ in population P, the linear variance, remainder variance, and twice their covariance are respectively 18.7%, 87.3%, and -6.0% of the observed variance. The covariance partly offsets the remainder, but the

linear variance is still only about one fifth of the observed variance. The same decomposition is calculated separately for Q.

B.5 Simulation uncertainty

For a rate estimate $\widehat{r}$ from M independent repetitions, its approximate Monte Carlo standard error is $\{\widehat{r}(1-\widehat{r})/M\}^{1/2}$. Rejection and calculation success rates use pointwise 95% Wilson intervals (Wilson, 1927). These intervals describe simulation uncertainty at fixed populations, not simultaneous coverage over every setting.

B.6 Component moment targets and uncertainty

The population first-order degrees of freedom $\nu_{P,\mathrm{FO}}$ in Equation (4.40) approximate the sampling-moment target $\nu_{P,\mathrm{mom}}$ in Equation (4.7). The target uses the actual sampling mean and variance of $\widehat{V}(P)$, measured over repeated samples from the same population at fixed sample size. Assuming positive $V(P)$, $\tau^2(P)$, and sampling variance, divide the two definitions and rearrange to obtain the exact identity

$$\frac{\nu_{P,\mathrm{FO}}}{\nu_{P,\mathrm{mom}}} = \frac{\left\{V(P)/\,\mathrm{E}\,\widehat{V}(P)\right\}^2}{\{\tau^2(P)/n_P\}/\,\mathrm{Var}\{\widehat{V}(P)\}}. \tag{B.9}$$

The numerator is the reciprocal of the squared mean ratio in Figure 6.1; the denominator is the middle-row predicted / observed sampling-variance ratio. A sampling mean $\mathrm{E}\{\widehat{V}(P)\}$ above $V(P)$ lowers the first-order degrees of freedom relative to the moment target when the variance prediction is otherwise accurate. The same calculations are made separately for Q.

Estimating the targets and their uncertainty.

For a component, let $X_s = \widehat{V}_s$ in repetition s. Use all finite values, including zeros, even if component degrees of freedom are undefined. From M repetitions, calculate

$$\bar{X} = \frac{1}{M}\sum_{s=1}^{M} X_s, \qquad S_X^2 = \frac{1}{M-1}\sum_{s=1}^{M}(X_s - \bar{X})^2. \tag{B.10}$$

For $S_X^2 > 0$, the estimated moment target is $2\bar{X}^2/S_X^2$. To evaluate Equation (B.9) for P, substitute $\bar{X}$ for $\mathrm{E}\{\widehat{V}(P)\}$ and S_X^2 for $\mathrm{Var}\{\widehat{V}(P)\}$.

The Monte Carlo standard error measures how much this estimated target would change if the whole set of M repetitions were repeated. Write $\mu = \mathrm{E}(X_s)$ and $v = \mathrm{Var}(X_s) > 0$, so the population moment target is $f(\mu, v) = 2\mu^2/v$. The delta method propagates the errors in $\bar{X}$ and S_X^2 through a first-order Taylor expansion. For changes h_μ and h_v, the general formula is

$$f(\mu + h_\mu, v + h_v) \approx f(\mu, v) + \frac{\partial f}{\partial \mu} h_\mu + \frac{\partial f}{\partial v} h_v. \tag{B.11}$$

The expansion point is (μ, v), the new values are $(\bar{X}, S_X^2)$, and the changes are $h_\mu = \bar{X} - \mu$ and $h_v = S_X^2 - v$. Since $\partial f/\partial \mu = 4\mu/v$ and $\partial f/\partial v = -2\mu^2/v^2$, substitution gives

$$\frac{2\bar{X}^2}{S_X^2} - \frac{2\mu^2}{v} \approx \frac{4\mu}{v}(\bar{X} - \mu) - \frac{2\mu^2}{v^2}(S_X^2 - v). \tag{B.12}$$

Both errors must be propagated jointly because the same repetitions estimate the mean and variance.

The mean error is exactly the average of $X_s - \mu$. To express the variance error in a similar form, expand $X_s - \bar{X} = (X_s - \mu) - (\bar{X} - \mu)$, square, and sum. The cross term uses $\sum_s (X_s - \mu) = M(\bar{X} - \mu)$, leaving

$$\frac{M-1}{M} S_X^2 = \frac{1}{M} \sum_{s=1}^{M} (X_s - \mu)^2 - (\bar{X} - \mu)^2. \tag{B.13}$$

Subtracting v and rearranging this identity separates the leading average from the two corrections:

$$S_X^2 - v = \frac{1}{M} \sum_{s=1}^{M} \{(X_s - \mu)^2 - v\} - (\bar{X} - \mu)^2 + \frac{S_X^2}{M}.$$

Since the repetitions are independent, $\mathrm{Var}(\bar{X}) = v/M$, so the standard deviation of the mean error $\bar{X} - \mu$ is $\sqrt{v/M}$. Its typical size therefore decreases as $M^{-1/2}$, and squaring it gives size M^{-1}. As M grows at fixed sample size, S_X^2 estimates the finite variance v, so the last term S_X^2/M also has size M^{-1}. Only finitely many count tables are possible at fixed sample size, so the centred squared deviations also have finite variance. Averaging these independent contributions gives the first term a standard deviation proportional to $M^{-1/2}$. The two corrections therefore become smaller relative to this leading sampling error. Omitting them at first order gives

$$S_X^2 - v \approx \frac{1}{M} \sum_{s=1}^{M} \{(X_s - \mu)^2 - v\}. \tag{B.14}$$

Substituting this expression and $\bar{X} - \mu = M^{-1} \sum_s (X_s - \mu)$ into Equation (B.12) gives

$$\frac{2\bar{X}^2}{S_X^2} - \frac{2\mu^2}{v} \approx \frac{1}{M} \sum_{s=1}^{M} \left[\frac{4\mu}{v}(X_s - \mu) - \frac{2\mu^2}{v^2}\{(X_s - \mu)^2 - v\} \right]. \tag{B.15}$$

The first term in brackets carries the mean error and the second carries the variance error. Replacing the unknown μ, v by $\bar{X}, S_X^2$ gives the estimated contribution from each repetition. Define

$$U_s = X_s - \bar{X}, \qquad W_s = (X_s - \bar{X})^2 - S_X^2. \tag{B.16}$$

The estimated contribution is

$$Z_s = \frac{4\bar{X}}{S_X^2} U_s - \frac{2\bar{X}^2}{(S_X^2)^2} W_s. \tag{B.17}$$

Equation (B.15) represents the target error as an average of independent repetition-level contributions. The standard deviation of their average is their individual standard deviation divided by $\sqrt{M}$. With $\bar{Z} = M^{-1} \sum_s Z_s$, its estimate is

$$\widehat{\mathrm{MCSE}} = \left\{ \frac{1}{M(M-1)} \sum_{s=1}^{M} (Z_s - \bar{Z})^2 \right\}^{1/2}. \tag{B.18}$$

The two terms within Z_s need not be independent: an unusually large X_s can affect both the estimated mean and variance. Calculating the standard deviation after combining the terms retains their covariance.

For the other diagnostic ratios, apply the same Taylor formula to $h(a, b) = a/b$, with derivatives $1/b$ and $-a/b^2$:

$$\frac{\widehat{a}}{\widehat{b}} - \frac{a}{b} \approx \frac{1}{b}(\widehat{a} - a) - \frac{a}{b^2}(\widehat{b} - b). \tag{B.19}$$

The mean ratio $\bar{X}/V(P)$ has a known denominator, so only the numerator error contributes. The population variance-prediction ratio $\{\tau^2(P)/n_P\}/S_X^2$ has a known numerator, so only the denominator error contributes. For the sample-based variance prediction, let $Y_s = \widehat{\tau}_s^2(P)/n_P$ and $\bar{Y} = M^{-1} \sum_s Y_s$. Both inputs to $\bar{Y}/S_X^2$ are estimated from the same repetitions, giving estimated contributions

$$Z_{s,\mathrm{ratio}} = \frac{Y_s - \bar{Y}}{S_X^2} - \frac{\bar{Y}}{(S_X^2)^2} W_s. \tag{B.20}$$

Use their sample standard deviation divided by $\sqrt{M}$, retaining the pairing between the numerator prediction and the squared deviation in each repetition. The ratio $\nu_{P,\mathrm{FO}}/\nu_{P,\mathrm{mom}}$ also has a known numerator, so its error follows by propagating only the estimated target's error through Equation (B.19).

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